



UNIVERSITY OF NAIROBI
DEPARTMENT OF COMPUTING AND INFORMATICS

**Applying Retrieval Augmented Generation(RAG) With LLM-Based
Chatbot To Enhance Patient-Facing Health Information Tools**

CCI - 599

Research Project
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P52/34980/2019

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Declaration

A research project submitted in partial fulfilment for the award of a **Master of Science degree in Computational Intelligence**. This project is my original work and to the best of my knowledge has not been presented anywhere for the purpose of an academic award.



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This project report has been submitted with my approval as the University Supervisor.



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Acknowledgement

I would like to appreciate the mentorship and role played by my supervisor, Dr. Eng. Muchemi among others not mentioned; for inspiring and imparting knowledge of artificial intelligence to me. This has given me the confidence and knowledge to pursue a modern research problem and the freedom to reason about other problems in the discipline of Computer Science.

Abstract

Large Language Models(LLMs) run on frozen parametric knowledge which limits their out-of-the box application for specific domains with constraints such as privacy regulations, proprietary data, costs, hallucination et cetera. They can be adapted to specific domains via prompt engineering, model fine tuning, alignment, Retrieval Augmented Generation(RAG) or a hybrid of either. Each of the techniques have their pros and cons; however, RAG has recently proven to be more cost effective for many general purpose applications. This study investigates the applicability of RAG for telemedicine chatbots by conducting experiments for information needs in telemedicine and creating a prototype chatbot based on RAG. We try to optimise various aspects of RAG such as combination of multiple retrieval sources with structured and semi-structured data formats like PDFs, website scrubs, Whatsapp chat exports as well as parquet datasets from Hugging Face. We also apply index optimisation approaches such as metadata enhancement and chunk size sizes. We evaluate the RAG pipeline using the RAGAS framework for core metrics including answer relevancy, context recall and faithfulness. The prototype achieves a high context recall rate of 56% and more than 75% in answer faithfulness and relevancy. The scarcity and quality of data proved to be a major limitation for this study; and a room for improvement for successful adoption of RAG in telemedicine.

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Abbreviations

LLM - Large Language Model

AGI - Artificial General Intelligence

RAG - Retrieval Augmented Generation

NLP - Natural Language Processing

CoT - Chain-of-Thought

RLHF - Reinforcement Learning with Human Feedback

GPT - Generative Pretrained Transformer

MIPS - Maximum Inner Product Search

MTEB - Massive Text Embeddings Benchmark

ETL - Extraction Transformation and Loading

GPU - Graphics Processing Unit

BERT - Bidirectional Encoder Representations from Transformers

DPO - Direct Preferences Optimization

TKO - Kahneman-Tversky Optimization

Chapter One

Introduction

Background

A prominent application of Large Language Models (LLMs) is their use for QA (Question Answering) in the format of conversational chats for wide-ranging tasks from general world knowledge queries, language translation, text summarisation, code generation et cetera. Patient navigation in the context of healthcare is a paradigm for coordinating integrated but inherently fragmented health services that often stand as a barrier to healthcare access. The goal is the consideration of a patient's individual needs to match them with the right services and providers (Budde et al., 2022). Patient navigators can be skilled professionals such as nurses, physicians or trained lay persons such as community health workers and former patients. Kenya has recently (Ministry of Health, 2023) launched a community health workers program to augment primary health care for all her citizens. Tele-health is similarly taking shape as populations seek online health services from online service providers; this necessitates a need for effective patient-facing health navigation tools that accords them a prompt and pleasant personal experience. The advent of LLM chatbot assistants may find application in the field of patient navigation when equipped with pertinent information to carry out patient navigation. Emergent LLM use-cases include tool co-pilots for various tasks such as coding and LLMs that can use tools (Schick et al., 2023). A natural use-case for LLM chats is in the field of customer service as bots and copilots. Tele-health is often served as a customer service function. Figure 1 in the next page summarises a landscape of use and tool chains developed for LLMs as per recent survey (Minaee et al., 2024). The survey revealed the general trend of current models including emergent use-cases as LLM agents that use tools and reference external knowledge bases to augment the static nature of LLM parametric knowledge. A maturing integration pattern for chatbots with LLMs is the use of RAG to adapt a particular domain's knowledge base to general purpose LLMs. The organisation of the components in these solutions are constantly

evolving majorly to cater for concerns about data privacy, costs and vendor choice. Crafting solutions from scratch that involve open-source LLMs is not easy for most organisations as there are various moving parts that need to be pieced together amid rapid evolution. Hosting custom LLMs is not a ready choice either due to their demands for specialised GPUs and computing resources that increase the cost of ownership. In the context of healthcare, sufficient guardrails is a key consideration against known LLM security vulnerabilities such as adversarial attacks, exposure of private patient or proprietary information and integrity of the generated responses.

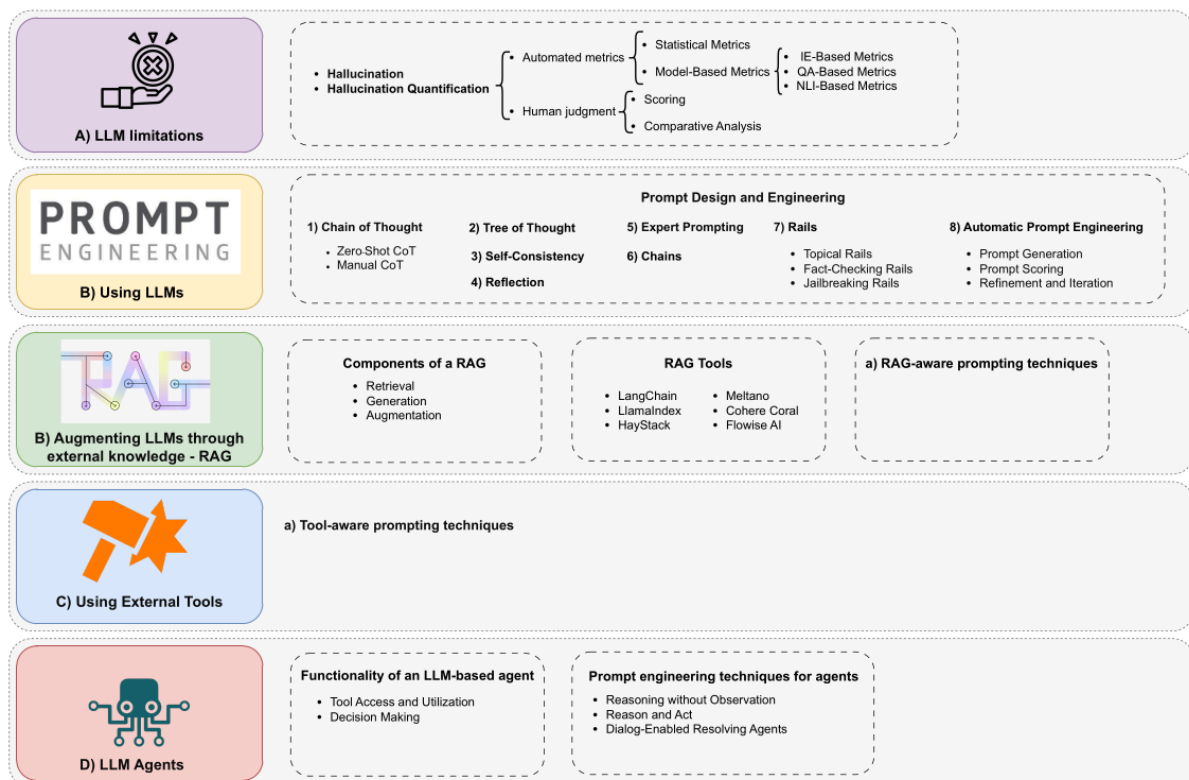


Fig 1. Survey of LLM use (Minaee et al., 2024)

Problem statement

Patient health care navigation is an increasingly important model for enhancing health care access as an increase of people seeking medical services meets an ever fragmented healthcare system. Tele-health delivers healthcare advice and services over telephone and internet communication tools including chat and email. The outcomes may include scheduling appointments for online or in-person visits with particular providers and specialists or complete advice for self-medication depending on the need at hand. LLMs have shown remarkable ability to conduct conversational interactions that fit in the format of telehealth; however, the limitations of transient state, variability of responses, hallucination and training on frozen information whose sources are often not disclosed for closed-source LLMs, make them not sufficient for out-of-the box application in specific domains. This study seeks to implement and evaluate a RAG-powered LLM chatbot by identifying relevant contextual information, transforming and tokenizing the sources for such information; such as telehealth chat data; particularly anonymized Whatsapp chats and existing private documents as knowledge sources. The study seeks to use LLaMA-2 Chat (Touvron et al, 2023) with GPT family, to create the chatbot with RAG to attend to telehealth queries and recommend potential tool use such as calendars for appointment booking, referral recommendations et cetera.

Objectives

General Objectives

To develop and evaluate a RAG pipeline over telehealth provider source knowledge base documents to enable an LLM-based chatbot to offer patient-facing health information navigation.

Specific Objectives

1. To identify the most critical and frequent health care information requested by patients from telehealth primary health care service providers
2. To develop an ETL pipeline to transform and index the data for purposes of RAG

3. To design a RAG model over an a Large Language Model to enhance patient navigation
4. To evaluate the effectiveness of the RAG model that assists in patient health navigation

Research Questions

1. What are the most relevant knowledge sources to extract the most frequently requested primary healthcare information for telehealth?
2. How do we extract and prepare the information from the knowledge sources for RAG source document embedding and indexing?
3. What is the optimal RAG pipeline over an open-source language model in a telehealth set-up with high demands for accuracy?
4. What are the metrics and factors needed to evaluate the quality of responses of a RAG-enabled LLM chatbot for patient navigation?

Justifications

LLMs are evolving rapidly and are emerging as a computing model therefore industry and academic learning is necessary to get the best of AI. As the Adyen team established; It requires more than an LLM to create useful chatbots for specific use cases(Hernández & Mora, 2023). There is much to be learned, methodologies and models to be derived from the best application of LLMs to real-world needs. This study is motivated by Penda Health, a primary healthcare-focused service provider whose suggestion on the need to develop more patient-facing health navigation tools echoes well with the current interest in NLP and AI in general. Penda Health generates more than 5000 chat interactions per day with people seeking health services over their Whatsapp channel.

Scope and Limitation

The research scope is limited to regular telehealth queries for Penda Health as a case study and does not seek to create an explicit knowledge-graph such as complete clinical advice or information for this context. The limitation on accessibility of proprietary data is a major hindrance. We shall seek to augment missing

information with synthetically generated questions. Accurate responses may demand specialised hardware resources limiting the scope of this study.

Assumptions

The research assumes availability of quality telehealth questions and knowledge base to meaningfully evaluate the resulting RAG pipeline. It assumes forthcoming cooperation with Penda Health for subject-matter expert consultation.

Chapter Two

Literature Review

Introduction

Patient navigation is a highly context-sensitive model for assisting health service seekers navigate the fragmented nature of health services. A specific problem; such as a disease, or disparity for a specific audience and nature of healthcare; such as primary versus specialty, community versus clinic service or insurance providers must be targeted precisely. Researchers in the journal of cancer education(Post et al., 2014) established the marked improvement in patient satisfaction and outcomes when human navigators targeted for this area are properly educated and therefore helpful for their patients. It therefore takes many different forms; offered by the array of personnel in the healthcare continuum.

Kenya launched the Community Health Care Promoters program (Ministry of Health, 2023) as a major preventive primary health intervention to address the disparity in provision of services to her citizens. This is an instance of patient navigation targeting primary health care for rural and urban residents utilising promoters as first contact for patients to link them up with the right service providers or outrightly offer basic primary care diagnostics. This study targets the applicability of LLM-based chatbots for online audiences that equally need to access tele-health services that are mainly powered by call centres accessed via chat applications. The chatbots need to be as good as human agents by demonstrating enough context-awareness of the health-provider. They must ground their responses on who or what is available, the different providers, their locations and capabilities. This effectively endows knowledge-intensive tasks on the chatbot as a patient navigator. LLMs are based on frozen parametric knowledge from the data they are trained on to power in-context and in-weight learning. RAG allows LLMS to perform non-parametric knowledge intensive tasks(Lewis et al., 2020). The next section gives a brief landscape and evolution of RAG, the identified gaps in the telehealth primary health-care domain as well as linking existing work to the proposed approach.

Existing work

Retrieval Augmented Generation

Large Language Models (LLMs) perform well for general purpose NLP tasks thanks in part to the massive data that endows them with in-context (Brown et al., 2020) and in-weight learning (Vaswani et al., 2017). This makes generative tasks much easier for LLMs but extractive tasks that require referencing temporal or recent information a challenge as it requires entire model retraining and finetuning. Meta researchers (Lewis et al., 2020) introduced RAG to bypass the need for retraining by proposing a means for LLMs to access external knowledge to supplement their frozen knowledge. They defined a retrieval and generation component over sequence-to-sequence models, now generally referred to as just LLM. They generalised the architecture as in Fig. 3 below.

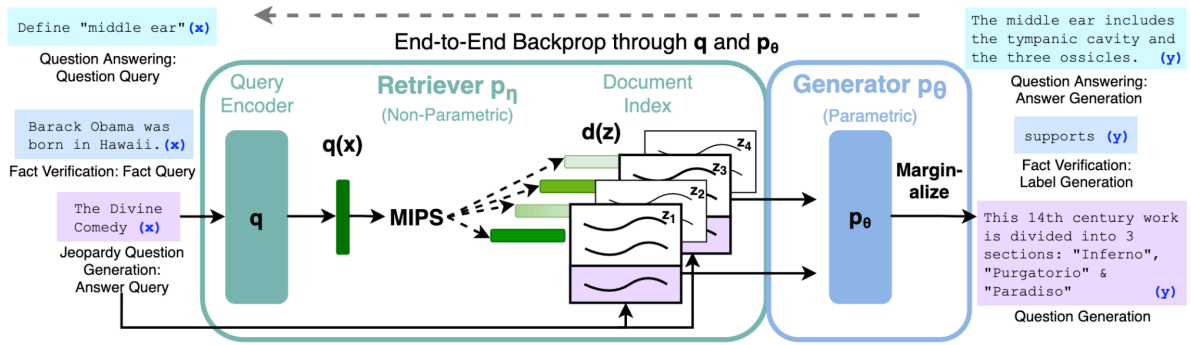


Fig 2. Retrieval Augmented Generation model (Lewis et al., 2020)

For input query x and query encoder $q(x)$ that applies a MIPS (Maximum Inner Product Search) function over x to retrieve a set of documents Z_i from the document index $d(z)$ using the top-k algorithm. The output Z_i is used as a latent input variable together with the original prompt against a pretrained model P_θ to give result y . The model above therefore generalises a retriever and generator as per the equations below:

$p_{\eta}(z|x) \leftarrow$ Retriever for η parameters returns *top-k* results given query x

$p_{\theta}(y_i|x, z, y_{1:i-1}) \leftarrow$ Generator considers the original query x , the retrieved results z and the latest tokens $i - 1$ to return result y .

The early RAG model proposed both a sequence-based and a token-based retriever. The sequence model took retrieved documents to generate the next tokens iteratively for each document to get the possible sources while the token-based model drew different documents for each token.

A recent survey of the state-of-the-art of RAG(Gao et al., 2023) reveals a comprehensive evolutionary path past the initial proposals of the Meta researchers(Lewis et al., 2020) and explores the current ecosystem of RAG. It reveals a research trend that mainly focuses on the optimization of RAG to address three main themes: *what to retrieve*, *when to retrieve* and *how to present the retrieved data*. The developments are also around the organisational components of RAG to achieve targeted objectives.

The earliest methodology gave rise to Naive RAG that comprises a straightforward linear progression of indexing, retrieval and generation. The indexing phase constitutes data-cleaning by transcoding from complex data formats such as HTML, XML or PDF to simpler formats such as markdown and plain-text. The pipeline would convert the simplified text to chunks to conform to LLM context-length limitations. The chunks would then be encoded into a vector format by an embedding model in preparation for the retrieval phase. The embeddings are therefore in key-value data stores to support efficient retrieval. The retrieval phase utilises the vector datastore to compute similarity scores from the encoded query using the top-k algorithm. The generation phase therefore sees an augmented query and conditions the model to elicit related responses.

The subsequent developments focused on the shortcomings of Naive RAG by targeting each of the stages; rudimentary indexing, poor retrieval and generation. Advanced RAG introduced intermediate optimizations such as index optimization by considering different data granularity and improving the quality of the data before indexing. The data would be augmented with state information such as timestamps, chapters and general metadata around the derived corpus. Retrieval phase in Advanced RAG is improved by introducing fine-tuning for the embedding model in

use and introducing dynamic embedding during pre-retrieval. The post-retrieval phase also includes re-ranking and prompt-compression to ensure relevance in generated output as well as conform to LLM context-length limitation.

Modular RAG succeeded Advanced RAG by adding flexible organisational flow and decomposing the stages into finer-grained modules that can be plugged to adapt to the task at hand. Figure below summarises the RAG paradigms(Gao et al., 2023).

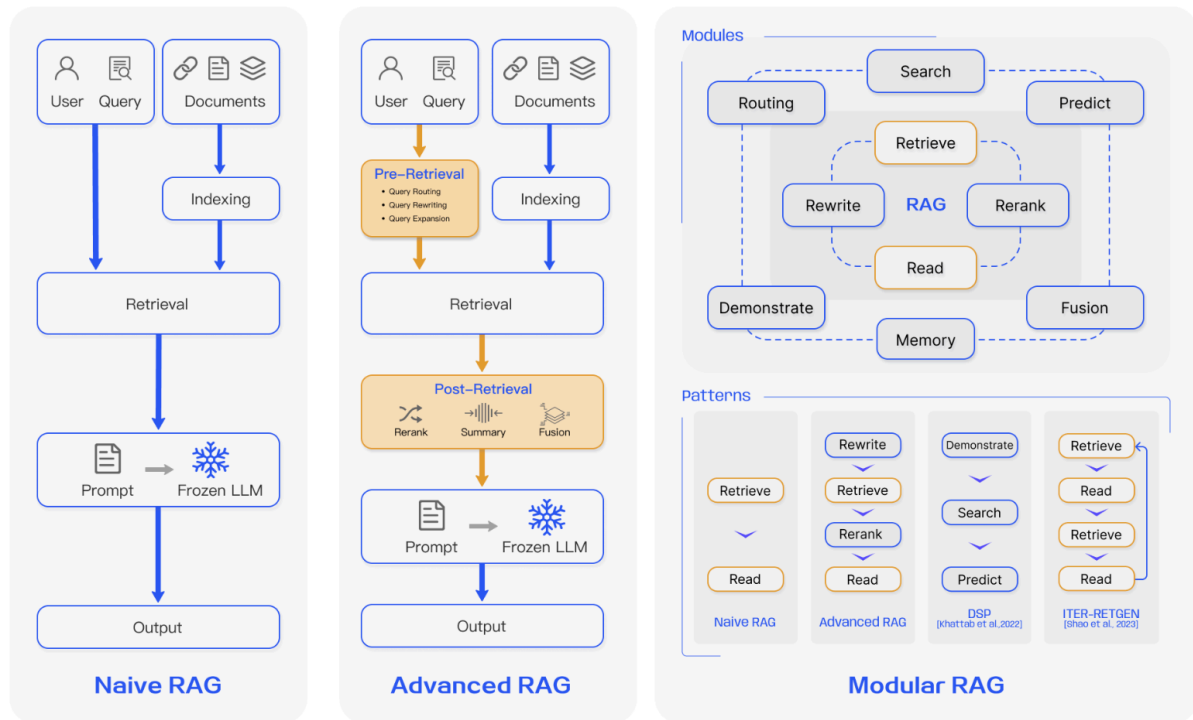


Fig 3. Retrieval Augmented Generation Paradigms (Gao et al., 2023)

The RAG ecosystem depicted above is fragmented and lacks general conventions for adoption for production-readiness. There are various language models, embedding models, indexing and data preparation techniques that vary vastly depending on the application at hand. RAG is therefore not a black box(Hernández & Mora, 2023) silver bullet to be plugged in a haphazard setting without significant consideration of the application domain.

Prompt Engineering

Large Language Model performance is heavily dependent on the quality of the prompts that they receive as the prompt serves as a conditioning to power in-context learning(Brown et al., 2020). Different styles of the same prompt can give dramatically different results, however labelling prompts strategically acts as helpful

demonstrations(Min et al., 2022) to guide the model in generating a desired response. Confoundingly, the labels and the demonstration data alone do not guarantee better responses but other aspects such as the distribution of the label space and the format of the conditioning data(Min et al., 2022). This is important in designing RAG systems because the ultimate pre-retrieval stages have to create proper prompts that include demonstration data derived from the documents. The granularity of the data, largely influenced by the chunking strategy and embedding model in-use(Gao et al., 2023) influences the distribution of the labels in the final prompt. The field of prompt engineering is rapidly evolving and has led to multi-pronged techniques for designing the ultimate prompts targeted for specific uses. Techniques in prompt engineering have alleviated a prominent limitation of LLMs; that while they perform well in text generation, they struggle with reasoning tasks. Google researchers(Wei et al., 2022) in their paper introduced the concept of *chain-of-thought*(CoT) prompting that encourages the LLM to break down a problem into multiple steps and reason methodically to generate responses. The figure below illustrates an example of the use of CoT.

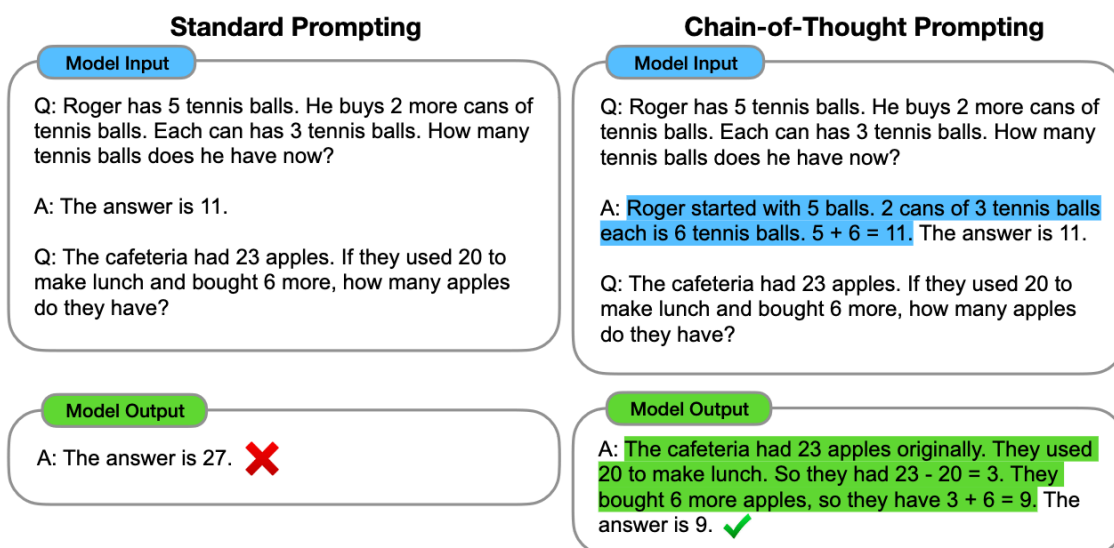


Fig 4. Standard Prompting vs Chain-of-Thought(CoT) prompting courtesy (Wei et al., 2022)

Many versions of CoT include *Zero-shot*(Kojima et al., 2022), *few-shot*(Chen et al., 2023), *manual-chain-of-thought* and *automatic-chain-of-thought*(Zhang et al., 2022) prompting. They vary mainly by the number of instructions needed to elicit a response or the technique of designing the prompt; either automatically or manually.

Fine Tuning and Alignment

Attention-based transformer language models(Vaswani et al., 2017) such as GPT family generalise well to different tasks with minimal finetuning unlike task specific architectures such as BERT(Devlin et al., 2018). Fine Tuning entails updating model weights by presenting a pre-trained model with supervised labels specific to a target task. The outcome is increased performance on the specific task at the expense of the need for massive amounts of such labelled data. There is however the danger of adopting spurious or wrong information from wrongly labelled data leading to generation of false information(Brown et al., 2020), popularly known as *hallucination*. Healthcare focused conversational AI might benefit from fine tuning because the language register can be terse unlike popular general knowledge datasets used on task-agnostic language models. Fine Tuning on clinical and medical journals will result in a more capable LLM for this study, however there's a huge cost related to labelling and tuning a pre-trained model.

Instruction Tuning

The need to elicit responses in a desired style and format is important for a task-specific conversational system based on an LLM. Instruction tuning is achieved by crafting prompts that specify the task the LLM is expected to achieve(Mishra et al., 2021). A dataset of positive and negative examples is used to train the LLM. Mishra et. al demonstrated that the resulting model achieves a marked improvement for new unseen tasks. The finetuning process can use self-generated fine-tuning instructions from the very LLM(Wang et al., 2022); termed *self-instruct* where Wang and team created a pipeline to generate instructions, input and output examples and proceed to discriminate invalid outputs for fine tuning the original model. They reported a 33% improvement over a base GPT3 model by using the self-instruct finetuning technique. [Figure 5](#) in the next page shows a high-level view of the pipeline.

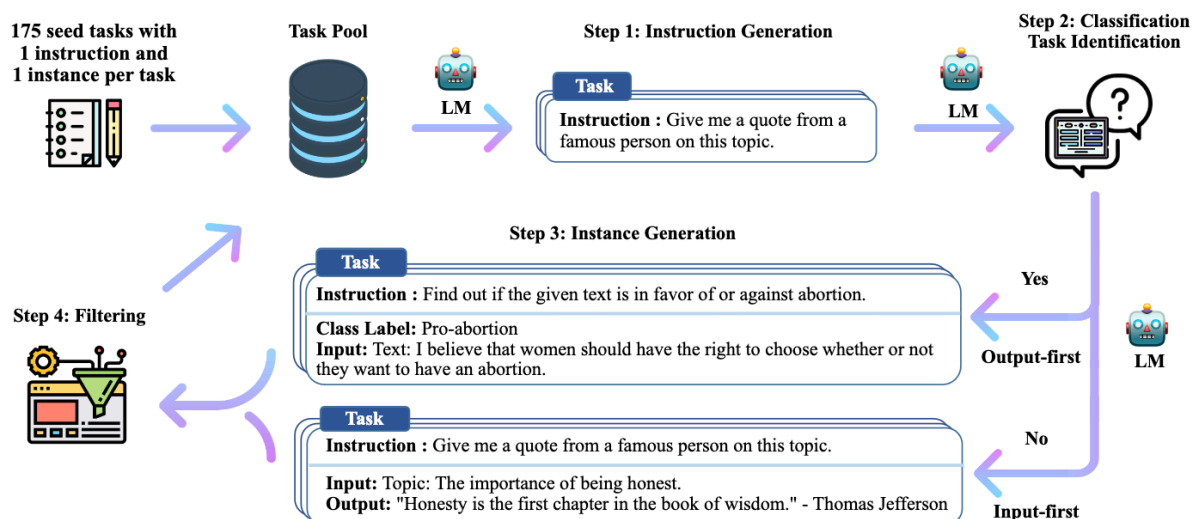


Fig 5. Self-Instruction fine tuning pipeline(Wang et al., 2022)

Alignment

Instruction tuning is one step at steering pretrained language models to desirable behaviour to address needs such as ethics and safety. Such needs have driven advancements in fine tuning, with approaches such as Reinforcement Learning with Human Feedback(RLHF) as described by OpenAI and Google researchers in their paper *Deep Reinforcement Learning with Human Preferences*(Christiano et al., 2017). They designed an algorithm to maximise a reward function from human feedback to power reinforcement learning. The schematic below summarises RLHF workflow.

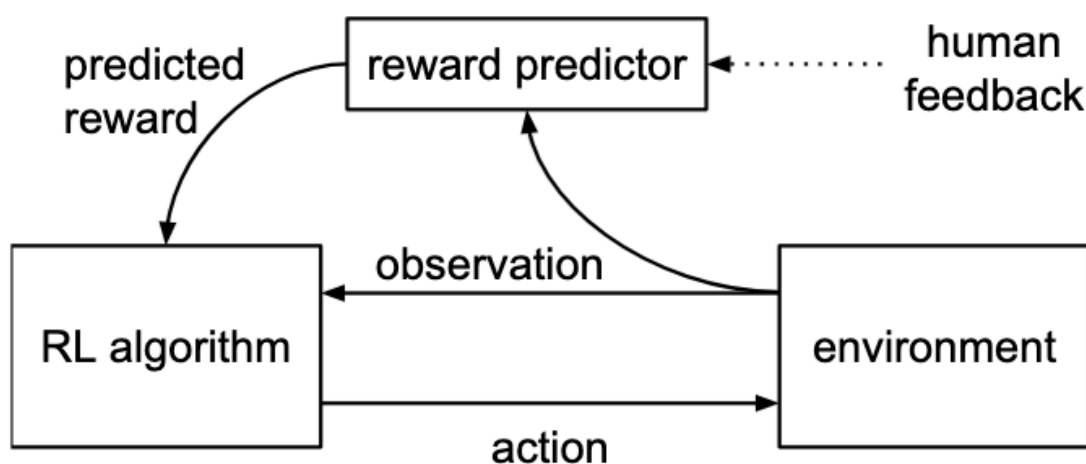


Fig 6. Reinforcement Learning with human Feedback(Christiano et al., 2017)

The technique RLHF has been leveraged to great effect to align current state-of-the-art task-agnostic models including GPT-4 and LLaMA albeit at a great cost as it requires thousands of human annotators to provide feedback on many datasets. **DPO**(Direct Preference Optimization) is a newer approach to alignment fine tuning deviating from the RLHF approach(Rafailov et al., 2023). The authors of DPO think that RLHF is too complex and unstable while DPO is computationally conservative as it optimises for human preferences while skipping the need for reinforcement learning altogether. Below diagram shows a comparison of RLHF and DPO.

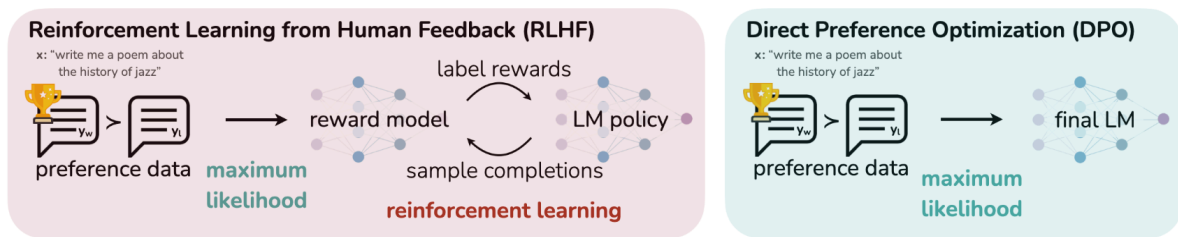


Fig 7. Schematic comparison of RLHF vs DPO(Rafailov et al., 2023)

A recent survey of the state-of-the-art of large language models(Minaee et al., 2024), also detailed a newer approach for LLM alignment termed **Kahneman-Tversky Optimization**, **KTO**(Ethayarajh et al., 2024) that unlike the above approaches does not require paired preference data(x, y_w, y_l) but only needs to know whether y is desirable or not given input, output (x, y) . KTO is based on *prospect theory* in economics proposed by Kahneman and Tversky in 1992 that defined the notion of *loss aversion*; the tendency for humans to be more sensitive to losses than gains of the same magnitude. The authors report that this method is much cheaper and superior to RLHF and DPO; especially for 1B to 30B-sized model parameters.

Patient Navigation

Patient Navigation is a broad term to describe the general process of a patient's journey through the health services continuum which is often fragmented or complex, resulting in barriers to successful health access(Honeycutt et al., 2013). It is therefore a patient-focused model for healthcare delivery that originated in the United States in 1995 for expanding access to cancer care for underserved populations in Harlem New York who were established to be majorly the underprivileged in society(Freeman, 2012). The programs and studies around patient navigation were

majorly focused on cancer patients to improve early diagnosis and care; Investigations on the effects on patient satisfaction outcomes(Post et al., 2014) established significant improvement in satisfaction for patients who were assigned navigators to help them through their cancer journey. This model has been expanded to other areas in health care delivery; yet Freeman Harold detailed the principles of successful patient navigation(2012), that clear demarcation for where it begins and ends for a given system of care must be outlined. There's a need to navigate patients from primary to tertiary sites and across many other boundaries including health care financing, post-treatment and end of life. The figure below shows a schematic of patient navigation transcending multiple boundaries.

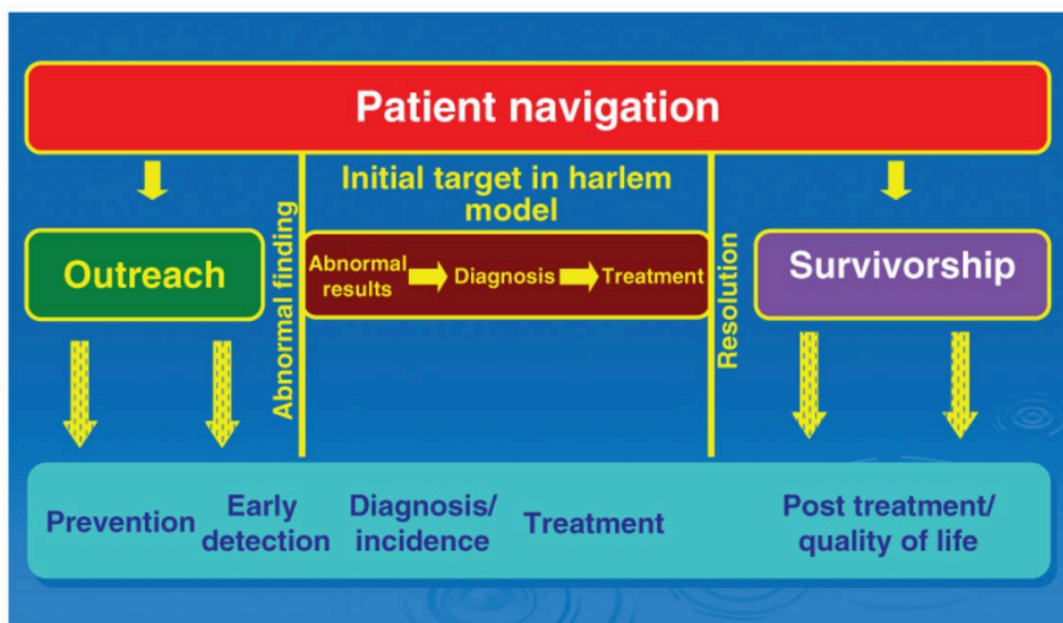


Fig 8. Patient Navigation Model(Freeman, 2012)

Patient Navigators

Personnel who help patients are traditionally trained medical personnel like nurses, doctors and clinical officers, lay persons or fellow patients in the case of cancer care(Post et al., 2014). They have knowledge of the healthcare system and procedures that help patients get what they need and thereby eliminate as many barriers as possible. Post and team identified the major barrier categories from their study on the effectiveness of patient navigation for cancer patients in New York. Below chart summarises the barriers.

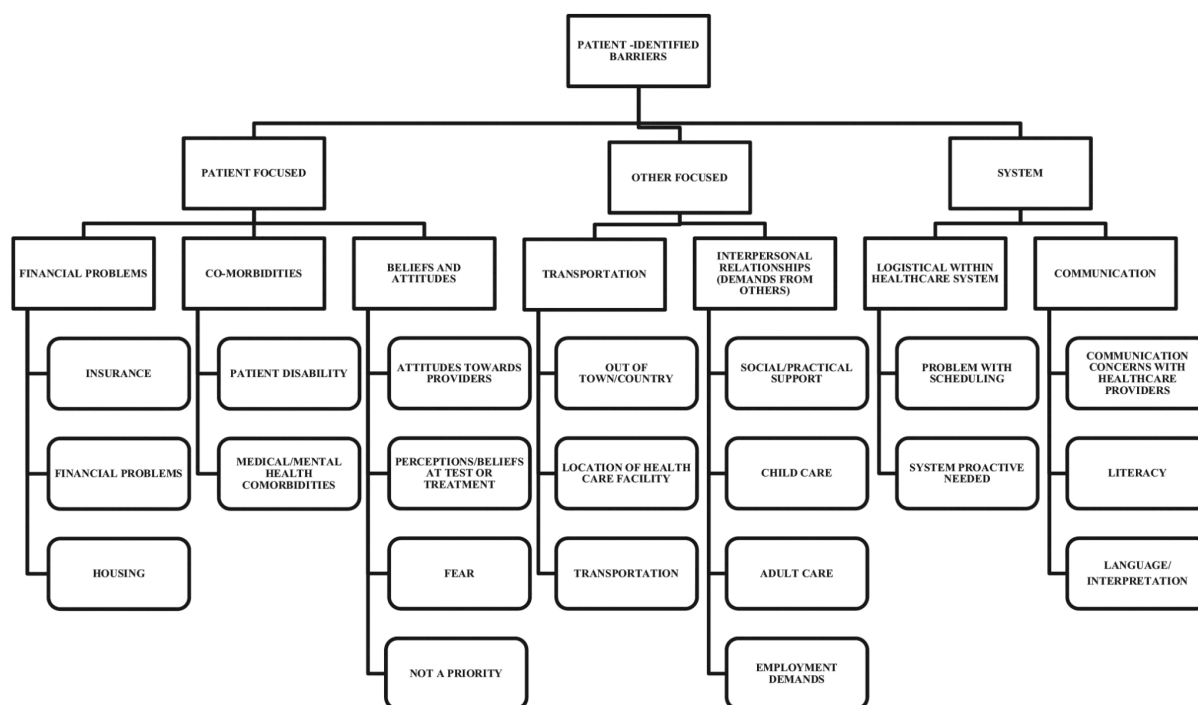


Fig 9. Identified patient-facing barrier categories(Post et al., 2014)

The chart above broadly categorised the barriers into patient focused issues such as finance, beliefs and attitudes; systemic issues such as logistical and communication among others. It is evident that by addressing these barriers, health care delivery can be dramatically improved without a need to expand existing healthcare facilities. In the next section we discuss the different modes of existing patient navigation.

Community Health Promoters

Kenya accelerated her focus on primary health care when it launched the community health promoters program(Ministry of Health, 2023) to act as champions for preventive public health care. The promoters are lightly trained non-medical personnel whose goal is to eliminate barriers to early diagnosis for rural and urban communities with the intended effect of relieving pressure from existing health facilities. They have basic knowledge to triage patients for simple illnesses, offer basic diagnostic tests and treatment as well as information on where to refer cases they are unable to handle. This is an effective champion model(Honeycutt et al., 2013) of patient navigation because a human touch and community ties bridge the beliefs and attitudes as well as systemic category of barriers detailed by Post(2014).

Telemedicine

Kenya has seen ubiquitous adoption of mobile telecommunication characterised by substantial internet penetration and advanced mobile money adoption. The foregoing has influenced the development and adoption of electronic health(eHealth) as detailed in a survey of the influence of design and implementation of mobile health interventions in Kenya(Sowon et al., 2022). Sowon focused on maternal health e-Health literature to understand the characteristics of the interventions and established that they mostly focused around the use of SMS technology to disseminate information. SMS was deployed mainly due to the practicality and reach of the technology for urban and peri-urban populations. Two-way SMS communications were manned by appointed personnel such as nurses to respond to maternal client queries. The study also noted the use of moderated WhatsApp groups(Patel et al., 2018) to guide mothers in postpartum phases. The unstructured nature of SMS and Whatsapp messaging helped to bridge technology literacy gaps as mothers would use natural language and expect responses in the same format.



Fig. 10. Kenya eHealth Policy Guiding Principles((Ministry of Health, 2016)

Kenya National eHealth Policy 2016-2030(Ministry of Health, 2016) identified the challenges and opportunities in e-health and among them; the fragmentation of technology leading to lack of interoperability and data exchange, insufficient synergy between players in the health industry as well as low technology literacy and high

cost of development. A notable characteristic is the transient nature of most interventions that were localised mainly around infectious diseases such as Malaria and maternal health in rural regions; many were short-lived pilots. Key among the guiding principles for the policy is the need for patient-centred healthcare services(see figure 11).

The advent of conversational AI is in-line with findings from Sowon et al.(2022) in that; maternal clients felt a better level of care when there was timely natural conversation and response with care personnel, yet were disappointed by the expected delays in responses which led to diminished use of the platforms. Moderated chat platforms(Patel et al., 2018) suffered similar challenges as response times and privacy issues. More recent mHealth solutions are offered by private providers, examples include: MyDawa¹, a licensed online pharmacy, SasaDoc² that offers telemedicine packages and laboratory services, Ponea Health³ is a concierge subscription service for linking patients with doctors and clinicians. Sophiebot⁴ is an AI chatbot that focuses on reproductive health question answering originally developed in Jomo Kenyatta University of Agriculture(JKUAT) as a student research project. Other providers are focused on health financing; commonly referred as insure-tech. Mtech⁵ is a player in aggregating various health insurance providers with supported facilities; making it easier to purchase policies and spend them. The fragmented landscape means a mobile application, a USSD code for each of them and many other technology infrastructures contributing to a high cost and cognitive load on users.

¹ mydawa.com

² sasadoctor.co.ke

³ Ponea Health

⁴ sophiebot.ai

⁵ mtek-services.com

Gap

Patient-facing tools for primary health service navigation is highly fragmented across the integrated health services continuum; getting information about where, when and who to go to for a given health need requires a significant journey through bureaucratic manned procedures. Health service providers should focus more on their core mandate and thereby make health more accessible by reducing unrelated cost structures such as maintaining huge call centres, help desks as well as offering cost-effective telemedicine with reduced physical visits. Specialised patient-facing health navigation tools such as USSD, mobile applications et cetera require significant investment by each provider while introducing fragmentation. They are also offered in user interactions that demand significant cognitive learning of each platform.

Linking existing work with proposed approach

This study proposes an intelligent chatbot approach to patient navigation; helping patients *navigate a fragmented* healthcare landscape. Significant progress and adoption in RAG(Retrieval Augmented Generation) over Large Language Models(LLMs) shows that humans effortlessly interact with computers in conversational natural language chat format.

Patient navigation according to Budde H, Williams GA, Scarpetti G, et al., (2022) improves access to healthcare without necessitating extra medical infrastructure by simply improving discovery and efficiency around the existing facilities.

Conceptual Model

This study envisions RAG application in a domain that has a strict demand for accurate responses, volatile information such as doctor and service availability and potentially distributed locations for in-person visits. We extract general information for the health providers from their websites by scrubbing relevant websites for general information, documents detailing their processes, available services and facilities as well as customer support data such as agent chat history from Whatsapp data export. Data cleaning and format standardisation shall follow for text tokenization and embedding using an open source embedding model. We shall

adopt a cache datastore to store embeddings generated by the model for efficient retrieval. The retriever shall perform pre-retrieval optimisation to extract top-k relevant documents and perform compression to limit the size of reference documents. The source document extracts that most closely match the input query are fused with the original query to generate the prompt that shall be fed to a pre-trained open-source LLM such as Ollama referencing LLaMA3. Figure 11 shows the conceptual model.

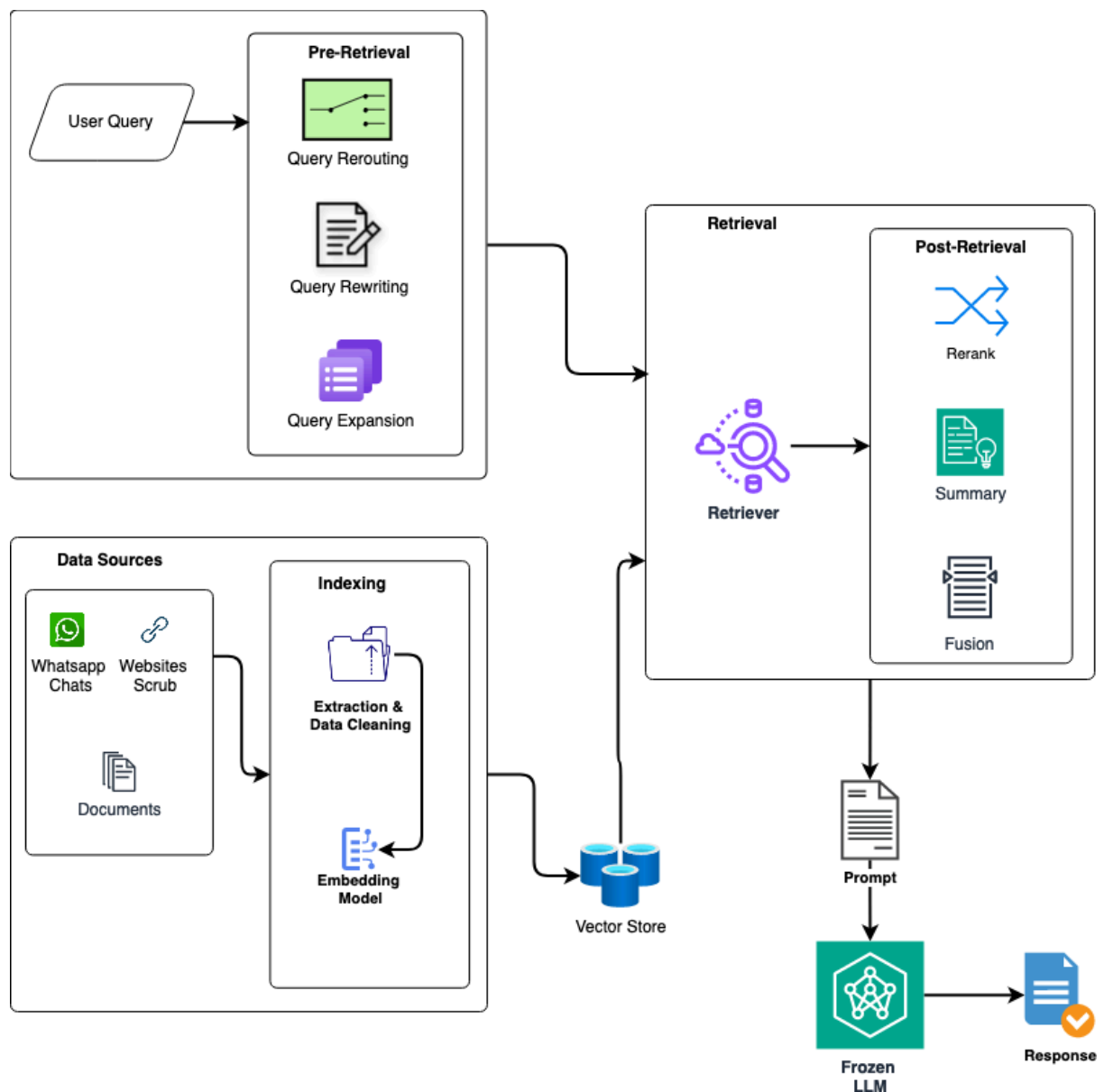


Fig 11. Conceptual Model for RAG pipeline

Chapter Three

Methodology

Introduction

This study runs two experiments to answer the research questions; first we conduct a requirements analysis phase via a brief onsite semi-structured direct interview at a case telemedicine provider, Penda Health⁶ in Nairobi to establish the most frequently sought information by their clients to identify knowledge sources to build an intelligent RAG-powered chatbot as a patient-facing healthcare navigation tool. This will yield insights such as the most frequent questions or category of questions, chief complaints and most common diagnoses to the respective complaints. We shall also identify the quality criteria for patient satisfaction for their existing virtual consultations conducted via manned chat channels.

The second experiment builds on the first by creating a RAG pipeline out of the information requirements established. We identify the varied knowledge sources such as structured and semi-structured documents, relevant web-links and acquire external datasets necessary to satisfy the information requirements of the chatbot. A prototype chatbot is then created to leverage this pipeline. We identify a suitable embedding model and a vector datastore. For the various sources, we identify a chunking and overlap strategy to optimise the vector indices as well as develop corresponding metadata for each chunk.

Finally we run and evaluate our prototype using the Ragas⁷ framework for each datastore; the goal is to measure the performance of our RAG pipeline which affects the performance of the chatbot.

⁶ [Penda Health](#)

⁷ [Ragas](#)

Research Design

This study combines interviews and design and creation strategy to answer the research questions. We create an instantiation in the form of RAG pipeline to realise an intelligent chatbot based on a frozen autoregressive transformer-based LLM.

Data Collection

We want to build our pipeline to continuously ingest, clean, chunk and index reference knowledge source documents from a primary health provider's repositories in an ETL process. The repositories include static websites, chat dumps, standard day-to-day operating documents such as duty rota, equipment, location and service inventory for each provider mainly in table formats. Publicly available medical questions datasets as a base information background for medical related queries is downloaded from Hugging Face repository. Evaluation datasets including synthetically generated data using Ragas from the base documents while the Hugging Face dataset shall be sliced and adapted for testing. It shall enable us to tune the parameters for chunking and metadata optimization. We leverage source *text-embedding-ada-002* embedding model, as it performs well on MTEB leaderboard⁸.

Data Preparation

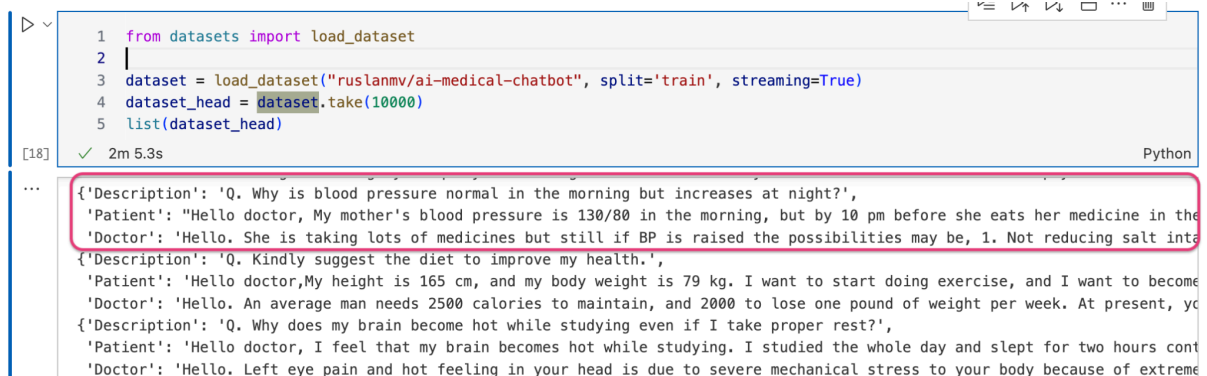
Document knowledge sources in PDF, semi structured and plain text are collected in a folder for ingestion via Langchain document loaders. Whatsapp history chats are exported and preprocessed lightly by removing names and personally identifying information. Figure 12 shows example preprocessing.

11:34 - This business uses a secure service from Meta	1 05/07/2024, 11:34 - This business uses a secure service from Meta
11:35 - Yunan: Hello Penda Health , Ephy here	2 05/07/2024, 11:35 - Patient: Hello Penda Health , Patient here
11:37 - Penda: Hello Ephy, Welcome to Penda Health.	3 05/07/2024, 11:37 - Penda: Hello Patient, Welcome to Penda Health
11:39 - Yunan: I'd like to have a consultation. What	4 05/07/2024, 11:39 - Patient: I'd like to have a consultation. What
11:48 - Penda: Great to hear that you'd like to have	5 05/07/2024, 11:48 - Penda: Great to hear that you'd like to have
11:49 - Yunan: I'd like to have a virtual consultant	6 05/07/2024, 11:49 - Patient: I'd like to have a virtual consultant
11:49 - Penda: Perfect choice, Ephy! The Virtual con	7 05/07/2024, 11:49 - Penda: Perfect choice, Patient! The Virtual c
13:02 - Penda: <Media omitted>	8 05/07/2024, 13:02 - Penda: <Media omitted>
13:02 - Penda: Kindly find attached lab request form	9 05/07/2024, 13:02 - Penda: Kindly find attached lab request form
	10

Fig 12. Light pre-processing for Whatsapp history chats.

⁸ [MTEB Leaderboard - a Hugging Face Space by mteb](#)

Dataset `ruslanmv/ai-medical-chatbot` from Hugging Face is streamed and partitioned. We acquire 10,000 triples of *description*, *patient* and *doctor* conversations out of the possible 250K such data for the purpose of this experiment.



```

1 from datasets import load_dataset
2
3 dataset = load_dataset("ruslanmv/ai-medical-chatbot", split='train', streaming=True)
4 dataset_head = dataset.take(10000)
5 list(dataset_head)

```

[18] ✓ 2m 5.3s Python

```

{'Description': 'Q. Why is blood pressure normal in the morning but increases at night?',
 'Patient': 'Hello doctor, My mother's blood pressure is 130/80 in the morning, but by 10 pm before she eats her medicine in the evening it rises to 160/90. What should I do?',
 'Doctor': 'Hello. She is taking lots of medicines but still if BP is raised the possibilities may be, 1. Not reducing salt intake, 2. Not taking medicine regularly, 3. Not exercising, 4. Not sleeping properly, 5. Not managing stress. I suggest you to consult a doctor for a proper checkup and treatment.'},
{'Description': 'Q. Kindly suggest the diet to improve my health.',
 'Patient': 'Hello doctor, My height is 165 cm, and my body weight is 79 kg. I want to start doing exercise, and I want to become a professional athlete. Can you suggest a diet plan for me?',
 'Doctor': 'Hello. An average man needs 2500 calories to maintain, and 2000 to lose one pound of weight per week. At present, you are consuming 2500 calories. I suggest you to reduce the intake to 2000 calories and start exercising. I will provide you a sample diet plan. You can consult a dietitian for a proper diet plan.'},
{'Description': 'Q. Why does my brain become hot while studying even if I take proper rest?',
 'Patient': 'Hello doctor, I feel that my brain becomes hot while studying. I studied the whole day and slept for two hours continuously. I feel very tired and my head is aching. What should I do?',
 'Doctor': 'Hello. Left eye pain and hot feeling in your head is due to severe mechanical stress to your body because of extreme fatigue. I suggest you to take a break and sleep for a few hours. You can consult a doctor for a proper checkup and treatment.'}

```

Fig 13. Hugging Face data acquisition and slicing.

Solution Design

The RAG pipeline is designed to assimilate different categories of document sources to culminate to multiple vector indexes to be accessed by our retriever in the general flow:

- For each document source S , apply a function that extracts to basic text by removing any special delimiters, markup or structuring with optimal chunk size C and favoured overlap O that preserves semantic meaning.

$$RAG\ sources(y) = \sum_{i=1}^n g(s_i | c_i, o_i)$$

- For each extracted source chunks y , generate embeddings with function h given embedding model m with supported context size. We attach a generated identifier and corresponding textual metadata for optimised retrieval and later referencing.

$$Embedding(e|m) = \sum_{i=1}^n h(y_i|m) + Id_i + text$$

- The embedding result is upserted to the vector store in batches. The pipeline is optimised to discriminate document sources that change frequently over the largely static ones to save costs and time.

This workflow is summarised by Figure 14 below.

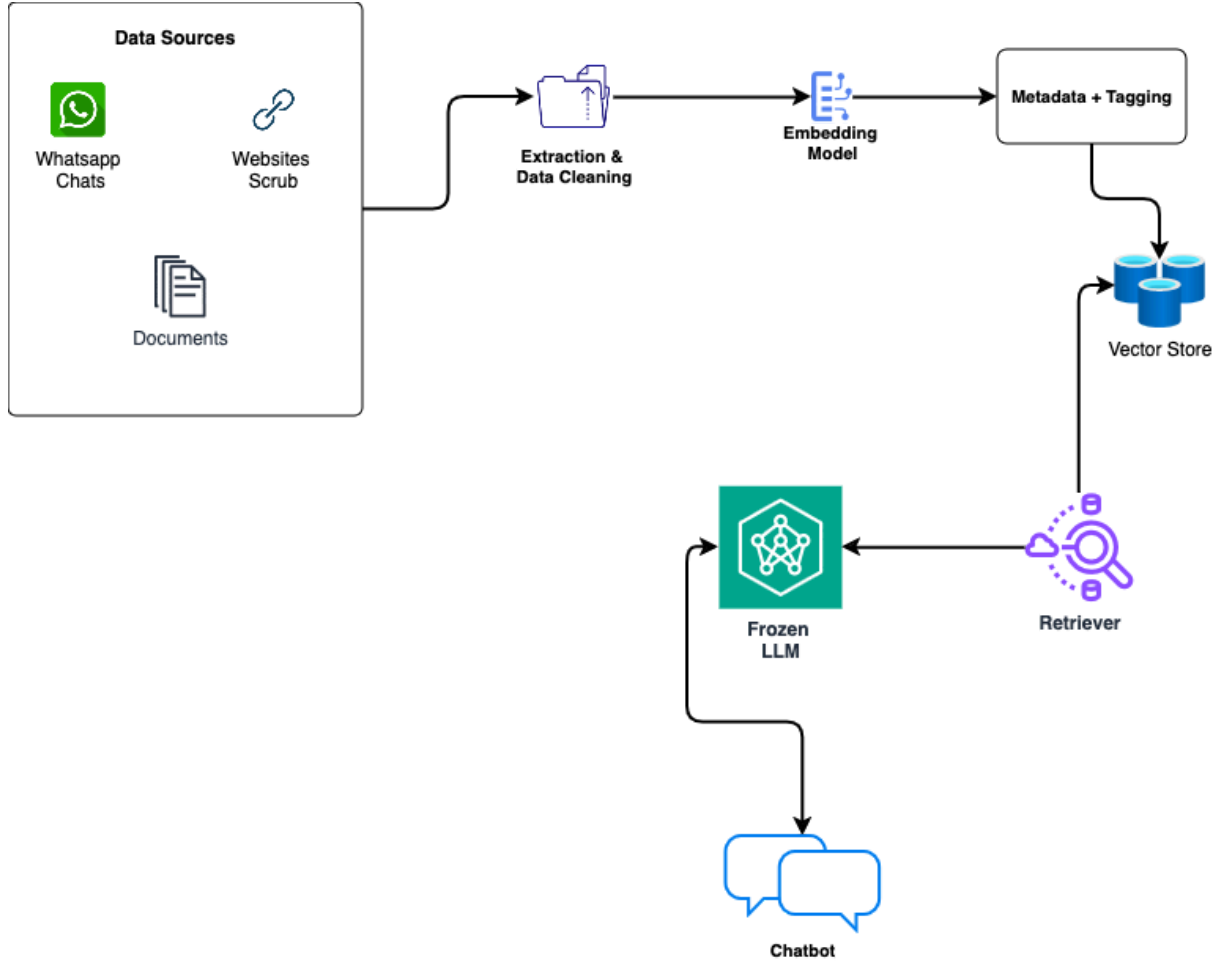


Fig 14. Solution Design

The retrieval phase of the design above exploits the RAG token model(Lewis et al., 2020) as it allows the generation phase of the retriever to aggregate multiple document sources. It is generalised by equation below:

$$p_{RAG} - Token(y|x) \approx \prod_i^N \sum_{z \in top-k(p(\cdot|x))} p_{\eta}(z|x) p_{\theta}(y_i|x, z, y_{1:i-1})$$

We have a choice of using the cosine similarity algorithm in place of the MIPS algorithm used by Lewis et al(2020) in the equation above. GPT 4.0 as p_{θ} is our frozen LLM that invokes the retriever p_{η} given query x . The LLM leverages on the retrieved contexts to iteratively generate responses to the chatbot. The chatbot interface provides a conversational interface and we use Flowise AI embeddable chat components on a simple HTML page. We have leveraged a managed vector

database by Pinecone⁹. It is also possible to use a free local file-based one like Facebook's FAISS¹⁰. We use the OpenAI embedding model text-embedding-ada-002. The RAG system features three major datasource formats including web documents in HTML, document material in PDF, plain text and markdown. We also have historical Whatsapp chats and each has a corresponding document loader to transform them into plain text formats. They serve as inputs to the embedding model which converts them into long vector strings annotated with metadata. The chunking sizes and overlaps are to be experimented to establish optimal configurations when running the experiments.

Running Experiment

Conducting Interview

The interview was carried out on July 12th 2024 at Westwood, Parklands Nairobi; Penda Health support office which domiciles their telemedicine facility. The [interview guide](#) is targeted to derive information such as demographics, nature of medical enquiries and their corresponding diagnoses.

Participants

We conveniently sampled Penda Health as a telemedicine provider. The centre has a group consisting of team leaders, clinical coordinators. Below table represents the distribution of the interviewees.

Category	Count	Role
Clinical Coordinators	2	Triage & first line contact
Providers	1	Trained medical personnel e.g doctors, clinicians
Supervisors	1	Quality assurance & leadership
Counsellors	1	Non-medical, gender and target groups counselling

Table 3: Interviewee category distribution

⁹ <https://www.pinecone.io/>

¹⁰ [Facebook AI Similarity Search\(FAISS\)](#)

Ethical Considerations

The privacy limitations associated with the area of study necessitated on-premise physical interviews with express permission from the management. Patient-doctor privacy and trade secrets of the organisation are in consideration. The [interview guide](#) in the appendix section details the specific questions and categories considered. Individual participants are each briefed on their rights during participation including consent to perform audio recording. The audio recordings are purely to be transcribed and thereafter discarded.

Running Prototype

The chatbot is built with Flowise AI for conversational interaction and showcase while a bare Python code version is built on a Jupyter notebook using Langchain LLM abstractions and makes it possible to evaluate our RAG setup performance using Ragas. This section details the experiment setup and runs using both approaches.

Document Store Setup

The outcome of the first experiment revealed the need for the chatbot to handle general inquiries and mostly reproductive health related queries. We set-up our prototype document stores with data sources from Penda Health website; specifically on their services, locations, and health tips. We also included sample PDF documents for reproductive health guides and sample Whatsapp chat history.

We experimented with two embedding models `text-embedding-ada-002` and `mxbai-embed-large` and opted for a managed vector database and a local one respectively. We opted for the former which has a bigger dimension(1536) compared to the later with 1024. We opted for a cosine similarity algorithm for generated vectors. The following documents were ingested using the above embeddings and the resultant vector output stored in the respective datastores.

```

1 #Load to Pandas frame
2 data = ds.to_pandas()
3
4 batch_size = 100
5
6 for i in tqdm(range(0, len(data), batch_size)):
7     i_end = min(len(data), i+batch_size)
8     # get batch of data
9     batch = data.iloc[i:i_end]
10    # generate embeddable text for each chunk
11    texts = [f"Description: {x['Description']}\n\nQuestion: {x['Patient']}\n\nAnswer: {x['Doctor']}" for _, x in batch.iterrows()]
12    # get Generate Unique IDs
13    ids = [ f"Q{i}" for i, _ in batch.iterrows()]
14    # embed text
15    embs = embeddings.embed_documents(texts)
16    # Create metadata to store in Pinecone alongside vectors
17    metadata = [
18        {'source': x['Description'],
19         'patient': x['Patient'],
20         'text': x['Doctor']}
21        for _, x in batch.iterrows()
22    ]
23    # add to Pinecone
24    index.upsert(vectors=zip(ids, embs, metadata))

```

✓ 44m 18.6s Python

Fig 15. Snippet to generate document chunks from ruslanmv/ai-medical-chatbot data in Pinecone
We created the embeddings from this dataset by concatenating as below in each row:

$$\text{Embedding} = \text{Description} + \text{Question} + \text{Doctor Response}$$

Each of the chunks are run through the embedding model to generate vectors. The vectors are stored alongside a generated ID and metadata; most importantly the **source** metadata. This optimises the retriever during inference and is also important when building specific retrieval chains. The image below shows a sample vector as stored in Pinecone data store. The unique identifier is also used during upsert operations.

1	ID	VALUES	
	Q2346	-0.00939011853, -0.0000802642098, 0.0330593176, -0.0355514847, -0.0347122848, 0.0217937212, -0.0044661867...	
SCORE	METADATA		
-0.0014	patient: "Hello doctor, I am 24 years old. I do not know what is wrong with me. I feel depressed all the time. I left home two years back because of my parent's be...		
	source: "Q. I feel depressed always with emotional instability. Please help to overcome it."		
	text: "Hi. The symptoms you have mentioned are symptoms of depression. You left home because of the behavior of your parents and then you are continuously f...		

Fig 16. Embedding lookup on Pinecone vector store

Table 4 shows documents uploaded for general information datastore.

Document/Source	Category	Document Loader	Chunking Processor
chat-na-penda-001.txt	Whatsapp chat	Langchain Whatsapp loader	Recursive text splitter

chat-na-penda-002.txt	Whatsapp chat	Langchain Whatsapp loader	Recursive text splitter
DoctorSchedulesPenda.pdf	Schedules	PDF loader	Markdown text
https://www.pendahealth.com	Locations, schedules, tips, services	Web Scrubber	HTML to markdown splitter

Table 4: Document categories for RAG setup

The chunking sizes and overlap are all 1000 and 200 characters respectively except for the Whatsapp chats which were only effective for 160 characters and an overlap of 50. Tuning these achieves different results. The screenshot in the next page shows a sample outcome of chunking doctor schedules represented in a tabular format in one of the document sources above.

Doctor Schedules at Penda Health

Doctor	Available days	Availability hours
Dr. Rob	Monday to Friday	8:00am to 12:00pm
Dr. Binocs	All days of the week	9:00am to 5:00pm
Dr. Sylvia	Monday, Tuesday, Friday, Saturday	Noon to 9:00pm

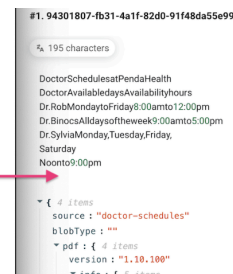


Fig 17. Chunking outcome for doctor availability at coarse granularity

The chunking in Fig. 10 generates exactly one chunk since the extracted content was much smaller than the defined size for 1000 thereby no truncation or overlap.

Chatbot Setup

Flowise creates a helpful chat component atop the retrieval chain as described in figure 18. This chatbot is embeddable in any HTML website. We use this interface to run and test the RAG pipeline for human consumption. A python script on a Jupyter notebook version references the same vector stores and LLMs and is suited for running evaluation for the setup in the next section.

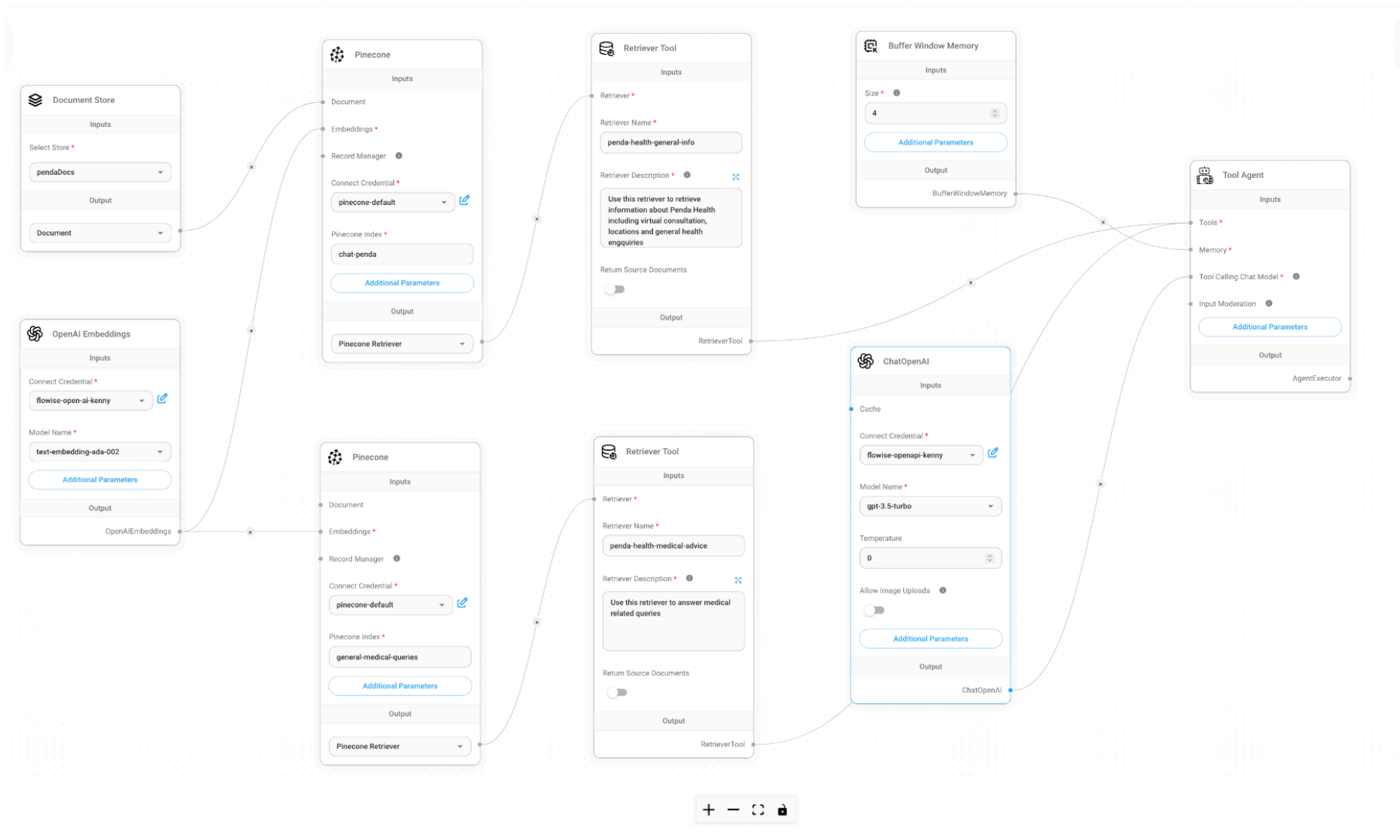


Fig 18. Chatflow setup for Flowise AI components

Prototype Evaluation

RAG applications require significant tuning due to the many moving parts in any one deployment. The prototype RAG production readiness and quality control is hard to track without rigorous formal testing. We leverage the RAGAS framework, embedding model and LLM for two main tasks: to generate triples of synthetic questions/answer/context from the base documents for evaluation. Secondly, to run evaluations against the pipeline for quantitative performance of the embeddings.

Metrics Framework

This study is interested in the below three main metrics described in the RAGAS paper(Es et al., 2023).

a. Faithfulness

$$F = \frac{|V|}{|S|}$$

Where $|V|$ is the number of statements that were supported by the recalled context according to the LLM and $|S|$ is the total number of statements. The value is a number between 0 and 1. The higher the figure, the more factually consistent the answer is.

b. Answer Relevance

$$AR = \frac{1}{n} \sum_{i=1}^n sim(q, q_i)$$

Where $sim(q, q_i)$ is the similarity of each of the original question against retrieved embeddings using cosine similarity algorithm. The value ranges between 0 and 1. The higher the value, the more directly the answer addresses the question without redundancies.

c. Context Recall

$$CR = \frac{\text{number of extracted sentences}}{\text{Total number of claims in ground truth}}$$

Where the extracted sentences from the ground truth that can be attributed to the context is divided by the number of claims in the ground truth. The value is between 0 and 1.

Evaluation Data Preparation

A set of *Question*, *Answer*, *Ground Truth* and *context* is required to evaluate the effectiveness of retrieval. The dataset `ruslanmv/ai-medical-chatbot` contains a triple of human annotated description of a patient's question, the query itself and a qualified doctor response. We slice 100 entries from this dataset and transform them to evaluation data as below:

Dataset Column	Evaluation Column
Description	Question
Doctor	Ground Truth
Answer	{to be generated by critic LLM}
Contexts	{Cosine similarity search on vector store}

Table 5: Evaluation Dataset derivation

Figure 19 in shows the light preprocessing of the dataset in our disposal for adaptation as evaluation dataset. We use GPT 3.5 to generate possible answers and create a Pandas dataframe.

```

1 import re
2 limit = 100
3 df = pd.DataFrame({
4     "question": [],
5     "contexts": [],
6     "answer": [],
7     "ground_truth": []
8 })
9 #Replace the initial 'Q.' in the description
10 pattern = r"~Q. "
11
12 for i, row in tqdm(enumerate(ds), total=limit):
13     if (i >= limit):
14         break
15     question = re.sub(pattern, "", row["Description"])
16     ground_truths = row["Doctor"]
17
18     answer= chain.invoke(question)
19     contexts = [docs.page_content for docs in retriever.invoke(question)]
20     df = pd.concat([df, pd.DataFrame({
21         "question": question,
22         "answer": answer,
23         "contexts": [contexts],
24         "ground_truth": ground_truths
25     })], ignore_index=True)
26 df

```

[101] ✓ 10m 3.0s Open 'df' in Data Wrangler Python

	question	contexts	answer	ground_truth
0	What does abutment of the nerve root mean?	[Hi. I have gone through your query with dilig...	Abutment of the nerve root refers to the compr...	Hi. I have gone through your query with dilige...
1	What should I do to reduce my weight gained du...	[Hi. You have really done well with the hypothe...	To reduce weight gained due to genetic hypothy...	Hi. You have really done well with the hypothy...
2	I have started to get lots of acne on my face,...	[Hi there Acne has multifactorial etiology. On...	Based on the information provided, it seems li...	Hi there Acne has multifactorial etiology. Onl...

Fig 19. Evaluation dataset derived from ruslanmv/ai-medical-chatbot

Running Evaluation

Once evaluation datasets; both adapted and generated are ready; it is a matter of selecting the metrics we are interested in from RAGAS. The LLM adopted for the RAG pipeline is interrogated automatically and the output accessible as a Pandas dataframe sampled below. The results are discussed in the next section.

...	question	contexts	answer	ground_truth	context_precision	context_recall	faithfulness	answer_relevancy
0	What does abutment of the nerve root mean?	[Hi. I have gone through your query with dilig...	Abutment of the nerve root refers to the compr...	Hi. I have gone through your query with dilig...	0.000000	0.000000	1.000000	0.857982
1	What should I do to reduce my weight gained du...	[Hi. You have really done well with the hypothe...	To reduce weight gained due to genetic hypothy...	Hi. You have really done well with the hypothe...	1.000000	0.457143	0.500000	0.969884
2	I have started to get lots of acne on my face,...	[Hi there Acne has multifactorial etiology. On...	Based on the information provided, it seems il...	Hi there Acne has multifactorial etiology. Onl...	0.805556	1.000000	0.500000	0.825156
3	Why do I have uncomfortable feeling between th...	[Hello. The popping and discomfort what you fe...	Based on the symptoms you have described, it l...	Hello. The popping and discomfort what you fel...	1.000000	1.000000	1.000000	0.823092

Fig 20. Sample RAGAS evaluation results

Limitations

Our proposals require use of general purpose embedding and language models; not specifically fine tuned for healthcare or localised for our purpose. Use of slang and Swahili language in some of our source data such as Whatsapp chats may not meaningfully enhance the performance of the proposed chatbot. We also intend to run this on consumer-grade hardware which is not ideal for autoregressive models that require massive parallel computing possible with specialised GPUs.

We also expect limited data from Penda Health as the actual data we might require for a full-fledged chatbot may be proprietary and limited by data privacy regulations.

Chapter Four

Results and Discussion

Introduction

This study sought to establish the applicability of an intelligent chatbot powered by an LLM in telemedicine setup. The study set out with four research questions. We review results of the first and second experiment as well as evaluation metrics in this section vis-a-vis the literature and conceptual framework.

Interview Results

The first research question used direct semi-structured interviews to understand the nature of the users of telemedicine for the case of Penda Health. They gladly provided summarised reports for three months preceding the interview day. The study established that more than half of the users are females ageing between 20 and 35 years. The graph in Fig. 21 illustrates the chief complaints recorded for the period between May and June 2024 for telemedicine visits. General consultations and reproductive health enquiries dominate followed by respiratory complaints.

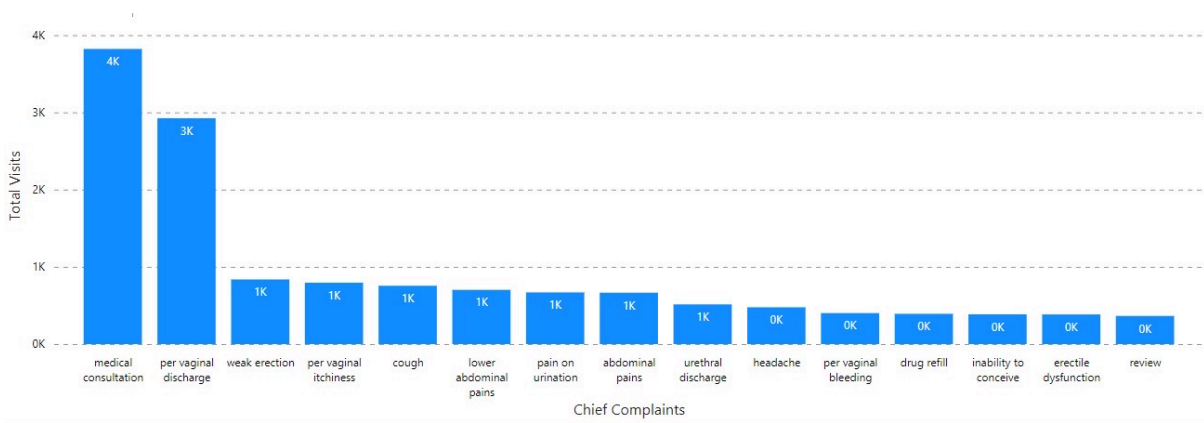


Fig 21. Telemedicine top chief complaints May - July 2024 (Courtesy Penda Health)

The chart below corresponds to the complaints in Fig. 21 as the diagnosis rendered by the Penda Health team in response to the complaints.

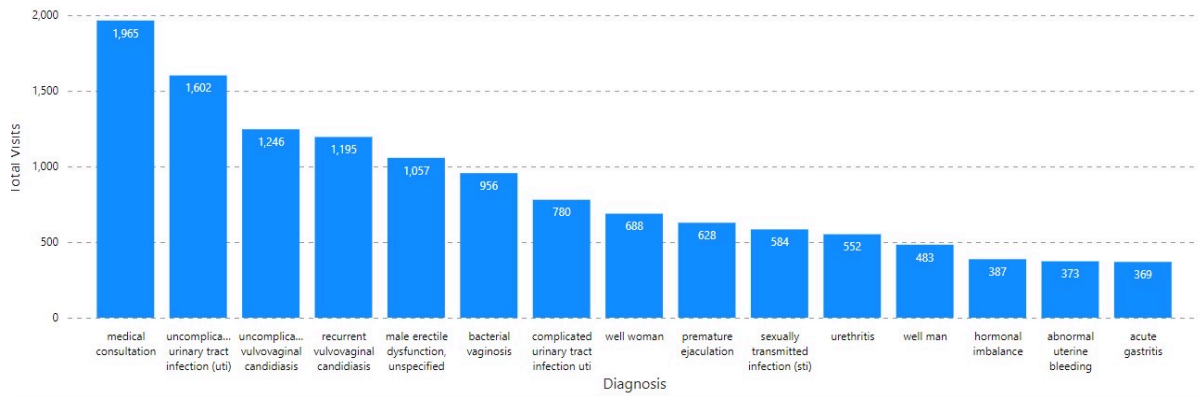


Fig 22. Telemedicine top diagnosis May - July 2024 (Courtesy Penda Health)

The direct interviews fetched insights from two clinical coordinators, a supervisor, a provider and a teenage counsellor. The demographic characteristics of users of telemedicine established that more than half of them, 64%, are female and they largely consult on general health inquiries and reproductive health, constituting more than 70% of the inquiries. These findings are in congruence with study on the Effects of Patient Navigation, Post et al (2014) that outlined the major barriers to patient navigation (summarised in [Fig 9](#)) that included beliefs/attitudes towards care providers and medical procedures generally. Women are more likely to seek medical attention than men and the telemedicine format reduces these barriers including fear and communication.

The study also sought to establish the quality assurance criteria used to measure satisfaction levels of users reaching them via chat. They reported that 94% of their patients rate their services as satisfactory. This is a qualitative self-assessed score given by the patients. This concurs with (Budde et al., 2022) findings on the outcomes for enhanced patient navigation; that marked improvement in satisfaction outcomes is observed with effective patient navigation. The fourth objective of this study tries to subject the chatbot to metrics such as faithfulness, context precision and answer relevancy using the Ragas (Es et al., 2023) framework.

Embedding Model Optimization

The study experimented with different chunking strategies for different document sources; we established failed recalls when chunking sizes exceeded 500 for whatsapp chats as the semantic meanings are lost due to short sentences and

answers. Fig. 25 shows cases when Whatsapp chats are not argued with metadata and chunking size set to 1000.

Documents with longer paragraphs resulted in relevant responses at chunk sizes of 1000 with overlaps of 200 characters; these included the website scrubs. Liu et al.(2023) in their study on the impact of significantly long contexts on LLM recall suggest that they lead to *lost in the middle* problem where useful information placed in the middle of a long context results in poor recall but perform better if placed at the beginning. Longer contexts may also lead to higher costs as most paid LLM providers charge per token(Es et al., 2023). Gao et al(2023) in their survey for RAG noted the importance of tuning the chunking strategies vis-a-vis metadata attachments to aid in attaching semantic meanings to different chunks while offering useful referencing tags. We incorporated metadata to optimise retrieval quality

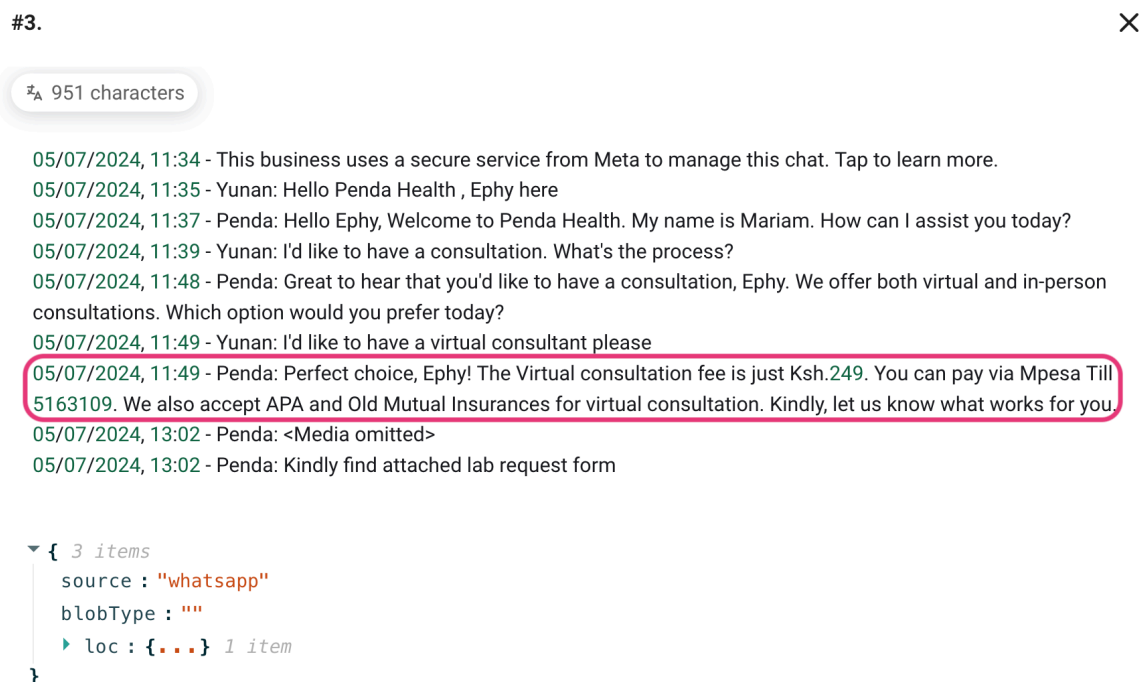


Fig 23. Sample chunking of Whatsapp history export using recursive text splitter

Fig. 24 shows sample chunking of a whatsapp historical chat at 160 characters. The results are evident in the responses below.

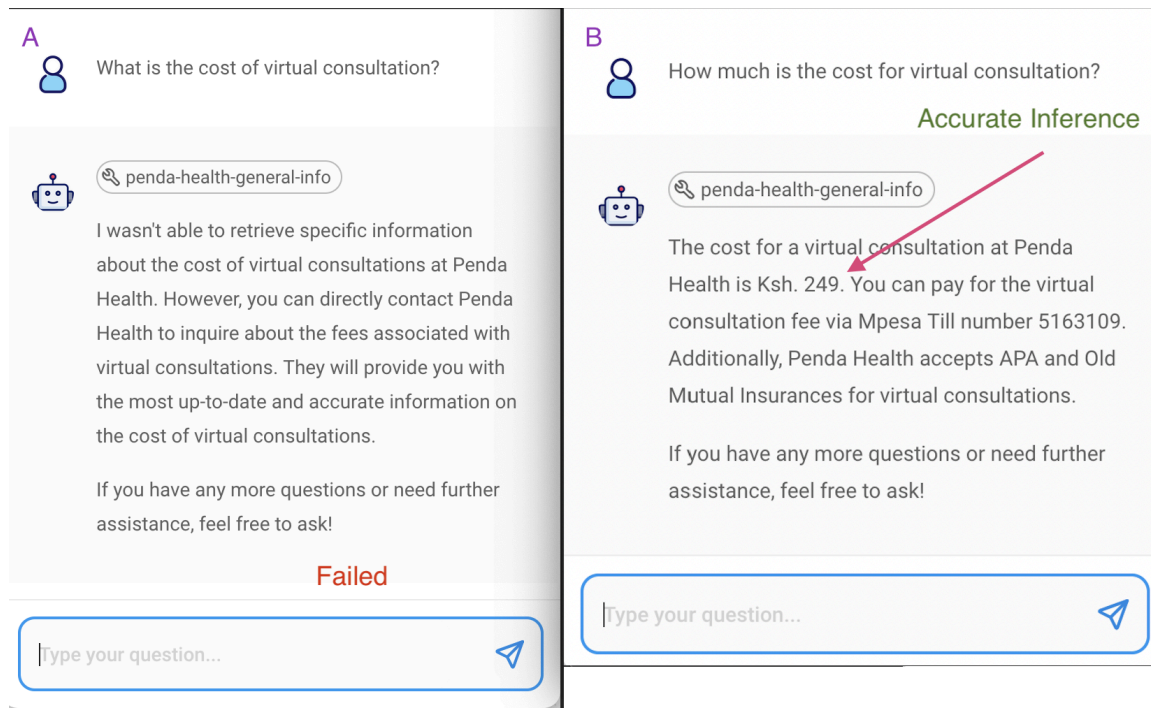


Fig 24. Sample chatbot response for different chunking configurations of whatsapp history

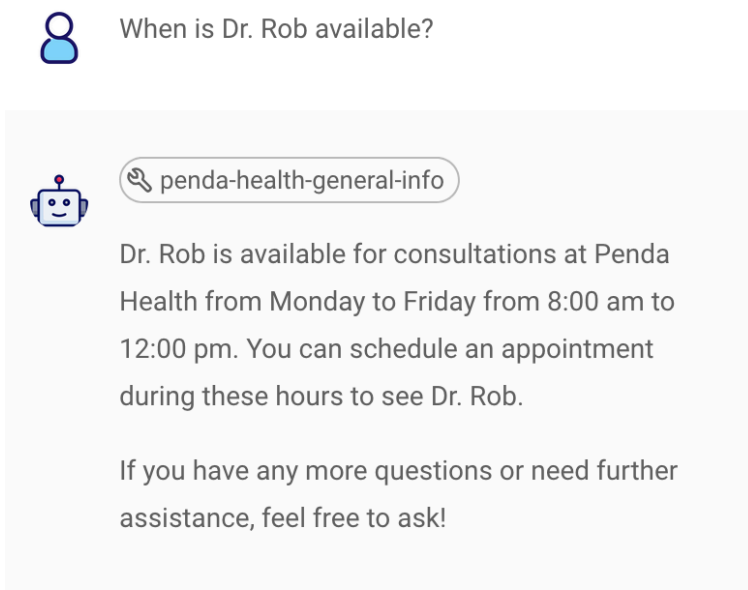


Fig 25. Sample chatbot response handling doctor availability

The documents can augment schedules such as doctor availability when presented in a table form. The conversation below shows a response to a query on the availability of a doctor.

Model Evaluation Results

The last question for this study sought to understand the measurement of the performance of the RAG pipeline created. The adopted approach was proposed by Es et al.(2023) as the RAGAS framework. We evaluated the prototype pipeline in three criteria discussed below. The data size is 100 questions against an indexed dataset of 10,000 triples of qualified annotation, patient question and doctor response.

Context Recall

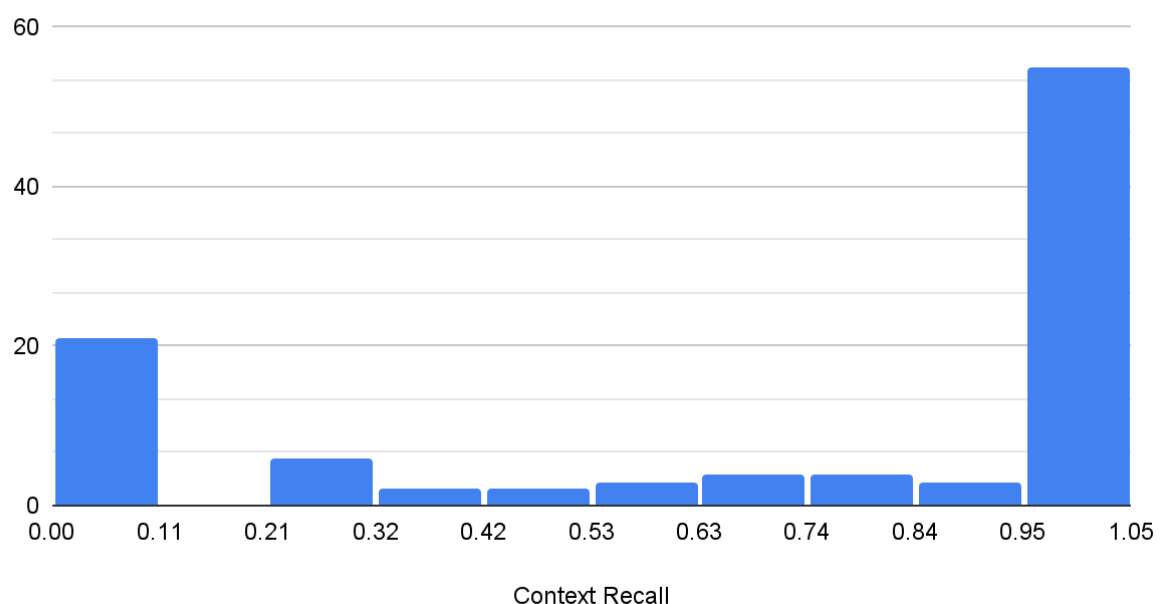


Chart 1. Histogram of prototype evaluation for Context recall

The chart above shows that our pipeline achieves a context recall rate of 56% for the sampled questions. The 20% poor recall rate is an indicator for insufficient ground truths in the documents. It is a majority of the questions where a doctor responds with a need for further clarification. In this case, the LLM appears to try to answer the question anyway without a backing from the knowledgebase. This is a sample ground truth:

Faithfulness

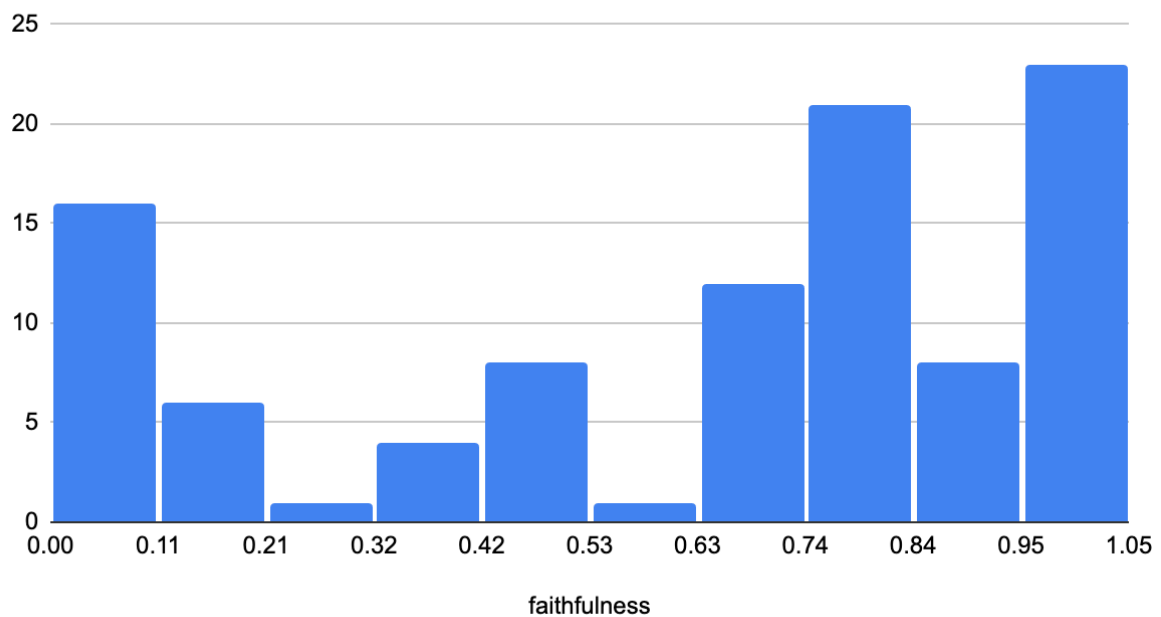


Chart 2. Histogram of prototype evaluation for Context faithfulness

The chart above shows the distribution of the prototype performance on the factual consistency of the answers generated in the sample. A majority of factually consistent answers are in the third percentile; implying lower hallucination rate (Es et al., 2023).

Answer Relevance

Results on answer relevance evaluation are summarised by Chart 3. Majority of the answers from our prototype, in the 75th percentile address the question directly without redundancies or irrelevant information.

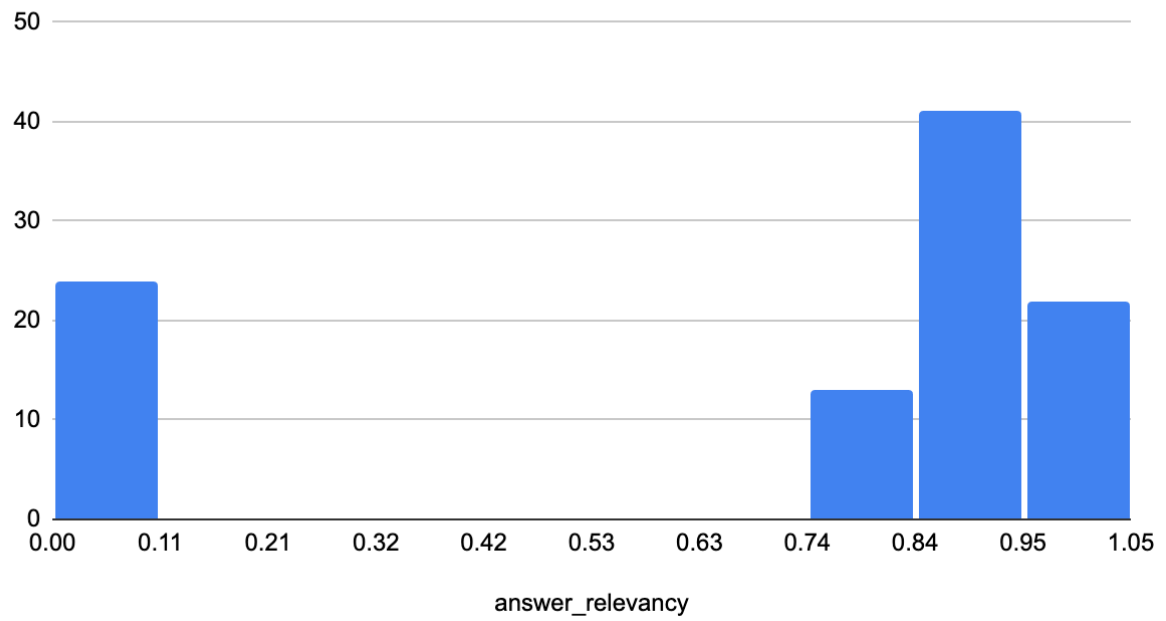


Chart 3. Histogram of prototype evaluation for Answer Relevance

Chapter Five

Conclusion and Recommendations

Introduction

This chapter takes a reflexive look at the study that sought to apply Retrieval Augmented Generation to power telemedicine Patient Navigation via an LLM-powered chatbot. We thematically retrospect through each of the results of the four objectives of this study and finally offer recommendations and appreciate the limitations.

Summary

We conducted direct interviews to gather requirements for telemedicine chatbots. and embarked on designing a RAG pipeline that would incorporate the most pertinent information for a chatbot to satisfy patient queries ranging from informational, reproductive health to basic medical advice. We experimented with embedding aspects such as models, chunking and overlap sizes to maximise the performance of our retriever to be able to handle general and medical related queries. Finally we explored the use of the RAGAS framework to evaluate the prototype given a *question*, *answer*, *context* and *ground truths* for three metrics: *context recall*, *faithfulness* and *answer relevancy*.

Identifying Relevant Telemedicine Information for patients

Patient navigation is at the heart of telemedicine solutions; insights gathered from interviewing Penda Health team reveal that they cater to address barriers of cost, privacy, convenience, beliefs and attitudes of segments of the population towards health care access. General information, reproductive health and counselling are the most important information sought by patients; while specific specialist examinations require specific information; often a physical visit or laboratory examination.

Developing ETL Pipeline for RAG

The insights from the first experiment informed the need for different categories of information required to furnish a RAG pipeline. There is a need for moderately static general information such as locations of medical facilities, catalogue of doctors and their specialisation, their schedules; pricing for different consultations, laboratory tests et cetera. Largely static information such as curated and well-known general health tips for the mostly requested information in reproductive health, respiratory illnesses et cetera. We explored the suitability of Whatsapp chat history from past patient interactions and achieved limited success with new challenges for anonymising patient information. We scrubbed Penda Health website for general information and locations; finally we acquired a publicly available general medical questions dataset that included *patient*, *doctor* and *description*; this contained 10,000 such questions.

Designing the RAG Prototype

The design of the RAG prototype was constrained by the requirements of the first experiment, the availability of data discussed in the foregoing subsection and largely by the costs of platform offerings for LLMs such as OpenAI, Pinecone vector store and hardware constraints for open source versions such as Llama3. We used Flowise AI to quickly prototype our chatbot as it abstracts away the components of RAG from different implementations. This allowed us to develop a document store wired to embedding models such as *text-embedding-ada-002* and a vector retriever connected to Pinecone vector store. We combined the separate types of data using a tool that interfaces with our LLM of choice; GPT 3.5 turbo and Llama3. This formed a retrieval chain that could be used by our chat model to comprise our chatbot. Local instances on commodity hardware of Llama3 proved impractical to generate embeddings for the 10,000 questions above and inference took as long as three to five minutes to generate responses.

Evaluating the RAG Prototype

We conclude that the evaluation techniques are strongly tied to the chunking framework and are quite straightforward. We generated synthetic questions for general information from Whatsapp historical chats, website scrubs and doctor

schedules using the RAGAS framework. Medical chatbot dataset from hugging face were adopted and a training dataset of 100 questions were evaluated on *context precision, context recall, faithfulness* and *answer relevancy*.

Contributions

This study has progressed our understanding on the process required to build a production-ready patient-focused intelligent chatbot to service primary health related queries. We have leveraged a general purpose LLM to serve specific responses to specific questions with a high context recall and answer relevancy required for the health services environment. Response turn-around times are much higher than manned chat-rooms; repetitive questions can be handled on a first shot basis, leaving the humans to deal with more pertinent cases only.

Limitations

This study relied purely on RAG techniques to create a chatbot for telemedicine setup. This means that general purpose LLM and embedding models were used without any finetuning and alignment strategies that would make the chatbot better positioned to handle core requirements such as patient-data privacy, avoidance of dispensing authoritative clinical advice without a human doctor examination et cetera.

The chatbot may be susceptible to adversarial attacks that could lead it to divulge patient private data if the historical whatsapp chats are not properly anonymised and sensitive data eliminated before utilisation as chatbot knowledge. It is possible to ask the chatbot what a given user asked about or what a doctor said about them.

The chatbot relies purely on a document retriever to generate responses; location-aware responses are not possible; for instance asking about directions to a healthcare facility or which one is nearest to a patient is a challenge for this implementation.

Future Work and Recommendations

There is much to build on the outcome of this study; both in practice and research. There is opportunity to explore the optimal techniques to adapt rapidly changing knowledge bases such as Whatsapp chats to argument document databases for RAG applications in telemedicine set-up. Specifically, how finetuned embedding models can be used to weed out names from chat participants and private information such as mobile-phone numbers, and any other Personally Identifying Information(PII). This can help improve the security of patient and organisational information.

Further research is required on LLM tooling for location information to help chatbots answer location-aware questions.

Practically, agent-based chatbots may be more suitable for telemedicine setups with a supervisor operation to moderate conversations for better experience for users as well as quality assurance on their performance.

Appendix

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Interview Guide

Introduction

Before the start of the interview, here are a few things to inform the interviewee about

- Let the interviewee know of your goals for this session
- Ask them for their frank opinion
- Let them know that there are no right or wrong answers
- Explain that the interaction is confidential
- Explain that the interaction is voluntary and they can stop anytime

Introduce yourself and Purpose

Briefly introduce yourself and explain the purpose of the interview.

"Thank you for taking the time to speak with me today. I'm conducting research to better understand Penda Health's services and patient experiences, both physical and virtual as part of a school research project"

Questions

Understanding Penda Health

1. What is Penda's mission & vision
2. What set's Penda apart from the rest

Understanding Their Patients

1. Can you describe the typical patients who use Penda Health services?
"What demographics (age, gender, occupation, etc.) do your patients usually belong to?" "Are there any common health concerns or conditions that they seek help for?"
2. How do patients usually find out about Penda Health?
"What channels or methods do they use to discover your services (e.g., referrals, online search, advertisements)?" "Which channels do they mostly reach Penda health through?"

Frequency of Service Usage

1. How often do patients use Penda Health services?
"Do patients tend to come back regularly for check-ups, or is it usually one-off visits?" "What are the typical intervals between their visits?"
2. Are there any trends in the frequency of visits for physical versus virtual consultations?
"Do patients use virtual consultations more frequently than physical ones, or vice versa?"

Preferences for Telehealth

1. Why do patients prefer to use telehealth services?
"What are the main reasons patients choose virtual consultations over physical visits?"
"Are there any specific advantages they mention, such as convenience, cost, or time-saving?"
2. Are there particular situations or conditions where patients prefer telehealth?
"For which health issues do patients most commonly opt for virtual consultations?"

Information Sought by Patients

1. What type of information do patients typically ask for during consultations?
"Are there common questions or concerns that come up frequently during both physical and virtual consultations?" *"Can you mention 3 most repetitive/common questions you have to answer?"*
"Do the information needs differ between initial consultations and follow-ups?"
2. How do you ensure patients receive accurate and comprehensive information?
"What systems or protocols are in place to handle patient inquiries efficiently?"
"Do you review the performance of your agents on how well they handled your patients?" *"If yes, what are the most important metrics or areas of evaluation?"*

Handling Virtual Requests

1. Who responds to patient requests when they reach out to Penda Health virtually?
"Can you describe the team or individuals responsible for managing virtual consultations and inquiries?" *"What qualifications or training do these staff members have to ensure high-quality care?"*
"Do you have manuals or reference materials for your agents to help answer patient queries?"
2. What is the typical process for handling virtual consultation requests?
"Can you walk me through the steps from the moment a patient reaches out to the conclusion of their virtual consultation?"
3. How do you manage the workload and ensure timely responses to virtual requests?
"Are there specific tools or systems you use to keep track of patient requests and responses?"

Conclusion

1. Thank you and next steps
"Thank you so much for your time and insights. This information will be invaluable for our research. Is there anything else you would

like to add that we haven't covered?"

At the end of the interview, remind the user about the goals of the interaction and what you plan to do with the information you have collected from them

Code Snippets

Evaluation Script

```
# %% [markdown]
# Setup the Environment by installing dependencies

# %%
%pip install -qU \
    langchain \
    langchain-community \
    langchain-openai \
    pinecone-datasets \
    pinecone-client \
    langchainhub \
    langchain-pinecone \
    datasets \
    pandas \
    ragas\
    python-dotenv \
    tqdm

%reload_ext autoreload
%autoreload 2

# %% [markdown]
# Setup credentials

# %%
import nest_asyncio
import os
import openai
from dotenv import load_dotenv
```

```

# Load environment variables from .env file
load_dotenv()
OPENAI_API_KEY = os.environ.get("OPENAI_API_KEY")
# Initialise the OpenAI API client
openai.api_key = OPENAI_API_KEY
nest_asyncio.apply()

# %% [markdown]
# Setu-up Datasets from Hugging face; we are using `ruslanmv/ai-medical-chatbot` to give
us enough medical questions. We shall use about `1000` questions from this set

# %%
from datasets import load_dataset

dataset = load_dataset("ruslanmv/ai-medical-chatbot", split='train', streaming=True)
dataset_head = dataset.take(10000)
list(dataset_head)

# %% [markdown]
# Setup Open AI embeddings model

# %%
from langchain_openai import OpenAIEmbeddings

model_name = 'text-embedding-ada-002'

embeddings = OpenAIEmbeddings(
    model=model_name,
    openai_api_key=OPENAI_API_KEY
)

# %% [markdown]
# Setup Pinecone Index for medical related questions from Hugging face

# %%
import time
from getpass import getpass
from pinecone import Pinecone, ServerlessSpec
from langchain_pinecone import PineconeVectorStore

index_name = "general-medical-queries"
pc = Pinecone(api_key=os.environ.get("PINECONE_API_KEY"))
if index_name not in pc.list_indexes().names():

```

```

pc.create_index(
    name=index_name,
    dimension=1536,
    metric="cosine",
    spec=ServerlessSpec(
        cloud="aws",
        region="us-east-1"
    )
)

while not pc.describe_index(index_name).status['ready']:
    time.sleep(1)

# connect to index
index = pc.Index(index_name)
time.sleep(1)
# view index stats
index.describe_index_stats()

# %% [markdown]
# Initialize Vector Store

# %%
from langchain_pinecone import PineconeVectorStore
text_field = "text"
vector_store = PineconeVectorStore(
    index, embeddings, text_field
)

# %% [markdown]
# Populate our index. Cache dataset to disk

# %%
import pandas as pd
from datasets import Dataset
from tqdm.auto import tqdm
from functools import partial

def gen_from_iterable_dataset(iterable_ds):
    yield from iterable_ds

ds = Dataset.from_generator(partial(gen_from_iterable_dataset, dataset_head),
    features=dataset_head.features)

```

```

# %% [markdown]
# Populate our Pinecone index by upserting the 10K rows from our dataset
`ruslanmv/ai-medical-chatbot`

# %%
#Load to Pandas frame
data = ds.to_pandas()

batch_size = 100

for i in tqdm(range(0, len(data), batch_size)):
    i_end = min(len(data), i+batch_size)
    # get batch of data
    batch = data.iloc[i:i_end]
    # generate embeddable text for each chunk
    texts = [f"Description: {x['Description']}\n\nQuestion: {x['Patient']}\n\nAnswer:
{x['Doctor']}" for _, x in batch.iterrows()]
    # get Generate Unique IDs
    ids = [ f"Q{i}" for i, _ in batch.iterrows()]
    # embed text
    embeds = embeddings.embed_documents(texts)
    # Create metadata to store in Pinecone alongside vectors
    metadata = [
        {'source': x['Description'],
        'patient': x['Patient'],
        'text' : x['Doctor']}
        for _, x in batch.iterrows()
    ]
    # add to Pinecone
    index.upsert(vectors=zip(ids, embeds, metadata))

# %% [markdown]
# Test The vector store by doing a similarity search

# %%
query = "Blood in stool"
vectorstore.similarity_search(
    query, # our search query
    k=3 # return 3 most relevant docs
)

# %%

```

```

from langchain_openai import ChatOpenAI

llm = ChatOpenAI(temperature=0)

# %% [markdown]
# Set Up a retrieval chain with prompt templates

# %%
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.output_parsers import StrOutputParser
from langchain_core.runnables import (
    RunnableParallel,
    RunnablePassthrough
)

prompt = ChatPromptTemplate.from_template("""
    You are a clinical coordinator at Penda Health, a telemedicine health provider.
    Question: {question}
    Context: {context}
    Answer:
""")
output_parser = StrOutputParser()
retriever = vectorstore.as_retriever()

retrieval = RunnableParallel(
    {"context": retriever, "question": RunnablePassthrough()}
)

chain = retrieval | prompt | llm | output_parser
# %%
import re
limit = 100
df = pd.DataFrame({
    "question": [],
    "contexts": [],
    "answer": [],
    "ground_truth": []
})
#Replace the initial `Q.` in the description
pattern = r"^Q. "

```

```

for i, row in tqdm(enumerate(ds), total=limit):
    if (i >= limit):
        break

    question = re.sub(pattern, "", row["Description"])
    ground_truths = row["Doctor"]

    answer= chain.invoke(question)
    contexts = [docs.page_content for docs in retriever.invoke(question)]
    df = pd.concat([df, pd.DataFrame({
        "question": question,
        "answer": answer,
        "contexts": [contexts],
        "ground_truth": ground_truths
    })], ignore_index=True)
df

# %% [markdown]
# Evaluate the medical dataset

# %%
from ragas import evaluate
from ragas.metrics import (
    faithfulness,
    answer_relevancy,
    context_recall,
    context_precision,
)

result = evaluate(llm=llm,
    dataset=Dataset.from_pandas(df),
    metrics=[
        context_precision,
        context_recall,
        faithfulness,
        answer_relevancy,
    ],
)

dt = result.to_pandas()

```